

Topological Criteria for Hypothesis Testing with Finite-Precision Measurements

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May 5, 2026

Abstract

We establish topological necessary and sufficient conditions under which a pair of statistical hypotheses can be consistently distinguished when i.i.d. observations are recorded only to finite precision. To accommodate finite-precision data, we introduce *finite-precision tests*: tests whose decision regions are open in the sample-space topology. We first show that, both for classical and finite-precision tests, the existence of such tests with finite-sample error control, asymptotic error control, or uniform convergence of the errors are all equivalent. A pair of null- and alternative hypotheses H_0 and H_1 admits a consistent finite-precision test if and only if both are F_σ in the weak topology on the space of probability measures $W := H_0 \cup H_1$. The hypotheses admit uniform error control under H_i if and only if H_i is closed in W , and admit uniformly consistent testing with *bounded* precision under metric separation of H_0 and H_1 . These criteria imply that, without regularity assumptions, conditional independence is not consistently testable from finite-precision data when the conditioning space has no isolated points — strengthening existing impossibility results to Polish sample spaces and showing that even pointwise consistency cannot be obtained. We introduce an equicontinuity assumption on the family of conditional distributions under which we recover consistent finite-precision testability of conditional independence with uniform error control under the null, provided sample spaces are Polish and the conditioning space is locally compact. The equicontinuity assumption is itself a finite-precision-testable hypothesis, so the resulting test for conditional independence is, in a precise sense, assumption-free.

1 Introduction

Whether a statistical hypothesis is testable remains an important question across the natural and social sciences. For example, judgements of conditional independence underlie many scientific inferences and are particularly fundamental in structural causal discovery (Spirtes et al., 1993). Recent results demonstrate that, unless regularity assumptions are made, any test for conditional independence with Type-I error control at level α has power at most α against every alternative (Shah and Peters, 2020). Nevertheless, it was believed that conditional independence is consistently testable (Györfi and Walk, 2012). The true situation turns out to be more complicated (Neykov et al., 2021). These recent results in conditional independence testing are arrived at via ingenious ad hoc arguments. Despite recent developments, there remains no simple, unified criterion for characterizing all and only the testable statistical hypotheses. In this respect, statistics is in sharp contrast to the theory of computation, where the testable hypotheses (co-semidecidable sets) receive elegant complexity-theoretic characterizations (Kelly, 1996). In this paper, we attempt to remedy this situation by laying out general topological conditions for statistical testability, generalizing results from Dembo and Peres (1994), Ermakov (2017), Genin and Kelly (2017) and Kleijn (2022).

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We demonstrate our results on the following elementary parametric examples since, contrary to many nonparametric hypotheses like conditional independence, their topological properties are evident. Let $X_1, \dots, X_n \sim \text{Bernoulli}(p)$ i.i.d. and consider for given $\varepsilon > 0$ the pairs of hypotheses

$$H_0 : p \in [0, 1] \cap \mathbb{Q} \quad H_1 : p \in [0, 1] \setminus \mathbb{Q}. \quad (1)$$

$$H_0 : p \in [0, 1] \cap \mathbb{Q} \quad H_1 : p \in [0, 1] \cap (\mathbb{Q} + \sqrt{2}) \quad (2)$$

$$H_0 : p \in [0, 1/2] \quad H_1 : p \in (1/2, 1] \quad (3)$$

$$H_0 : p \in [0, 1/2) \quad H_1 : p \in (1/2, 1] \quad (4)$$

$$H_0 : p \in [0, 1/2 - \varepsilon) \quad H_1 : p \in (1/2 + \varepsilon, 1] \quad (5)$$

As might be intuitively clear, for these pairs of hypotheses one can achieve different consistency properties of the test, ranging from uniform consistency for (5) to mere consistency for (2), and (1) is not consistently testable. The various notions of consistency and error control are made precise in Section 2, where we also show that existence of tests with many seemingly different formulations are in fact equivalent (Theorem 2). Our main results, presented in Section 3, state that these modes of testability are characterised by the topological properties of the hypotheses: whether they are F_σ (i.e. a countable union of closed sets), closed or open, or clopen with respect to the subspace topology on $W := H_0 \cup H_1$, or metrically separated. We don't restrict to parametric hypotheses, but we consider arbitrary nonparametric hypotheses H_0 and H_1 as subsets of the space of Borel probability measures $\mathcal{P}(\mathcal{X})$ on a separable metric space $\mathcal{X}$. The topological characterisations are with respect to the weak topology on $\mathcal{P}(\mathcal{X})$. The data is assumed to be i.i.d. from a single $\mathbb{P}$ in either H_0 or H_1 . Notably, most of our results do not require any regularity conditions on the probability measures under consideration. However, it turns out that the weak topology characterises testability with *some* type of regularity: that the critical regions of the tests are open — a property that is useful when considering finite-precision measurements, to be further motivated in Section 1.1 below.

Our main application of these topological characterisations is to analyse the feasibility of conditional independence testing. For given sample spaces $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ and set of probability measures $W \subseteq \mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ we consider

$$H_0 := \{\mathbb{P} \in W : X \perp\!\!\!\perp_{\mathbb{P}} Y \mid Z\} \quad H_1 := \{\mathbb{P} \in W : X \not\perp\!\!\!\perp_{\mathbb{P}} Y \mid Z\}, \quad (6)$$

where conditional independence, denoted with $X \perp\!\!\!\perp_{\mathbb{P}} Y \mid Z$, means that for all measurable A and B the factorisation $\mathbb{P}(X \in A, Y \in B \mid Z) = \mathbb{P}(X \in A \mid Z)\mathbb{P}(Y \in B \mid Z)$ holds $\mathbb{P}(Z)$ -almost surely. Conditional dependence is denoted with $X \not\perp\!\!\!\perp_{\mathbb{P}} Y \mid Z$. In Section 4, we show that conditional independence and conditional dependence are both dense in the weak topology on $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$. It follows from the Baire category theorem that there does not exist a consistent FP-test for conditional independence when $W = \mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$. Hence, conditional independence testing requires assumptions, that is, a restriction of the set W . We introduce the space of distributions whose conditional distributions $\mathbb{P}(X \mid Z)$ or $\mathbb{P}(Y \mid Z)$ are equicontinuous, and investigate some properties in Section 5. We show that this space is closed in the weak topology, and that conditional independence is closed in this space. In Section 6 we discuss the consequences for various modes of testability of conditional independence.

1.1 Accommodating finite-precision measurements

Given a test $\varphi_n : \mathcal{X}^n \rightarrow \{0, 1\}$ and a sample $x \in \mathcal{X}^n$, the verdict of the test is given by the evaluation $\varphi_n(x)$. From now on, we write $\{\varphi_n = i\}$ for the region $\{x \in \mathcal{X}^n : \varphi_n(x) = i\}$. In standard presentations of hypothesis testing, the acceptance and rejection regions of a test are taken to be arbitrary *measurable* sets. But is it really natural to consider a test that rejects if the sample point is rational-valued? Or if it is precisely π ? If measurements of real-valued quantities can be made with only finite precision, such tests cannot be implemented.

We adopt the following idealisation. A measurement device, given a true sample value $x \in \mathcal{X}^n$, reports a truncated value $x' \in \mathcal{X}^n$ and a precision parameter $\varepsilon > 0$ with $d_\infty(x, x') < \varepsilon$, where d_∞ denotes the ℓ^∞ product metric on $\mathcal{X}^n$. The empiricist observes the pair (x', ε) but *not* the true value x ; what she knows after a single measurement is that $x \in B(x', \varepsilon)$, the open ball of radius ε around the

report. By requesting reports at decreasing precision $\varepsilon \downarrow 0$, the empiricist can in principle pin down x to arbitrary accuracy.

Given a measurement (x', ε) and a test φ_n , the empiricist declares verdict $i \in \{0, 1\}$ if $B(x', \varepsilon) \subseteq \{\varphi_n = i\}$, in which case $\varphi_n(x) = i$ regardless of where x sits inside $B(x', \varepsilon)$. If no such i exists, the empiricist refines the precision and tries again. For this protocol to terminate at samples lying in $\{\varphi_n = 0\}$ or $\{\varphi_n = 1\}$ we require these critical regions to be open in the topology on $\mathcal{X}^n$: openness of $\{\varphi_n = i\}$ guarantees that for any x in this critical region, sufficiently small ε yields $B(x', \varepsilon) \subseteq \{\varphi_n = i\}$. If $\mathcal{X}^n$ is connected, $\{\varphi_n = 0\}$ and $\{\varphi_n = 1\}$ cannot both be nonempty, open, and cover the entire sample space; we therefore introduce the region $\{\varphi_n = 2\}$ that recommends neither acceptance nor rejection of H_0 , but suspension of judgement.¹

Definition 1. A *finite-precision test (FP-test)* is a measurable map $\varphi_n : \mathcal{X}^n \rightarrow \{0, 1, 2\}$ such that $\{\varphi_n = 0\}$ and $\{\varphi_n = 1\}$ are open.

Abusing terminology, we sometimes refer to a *(FP-)testing sequence* $\varphi = (\varphi_n)_{n \in \mathbb{N}}$ as a *(FP-)test*. For an FP-test, the output protocol terminates with the correct verdict whenever x lies in the interior of any of the three regions; only at the boundary of $\{\varphi_n = 2\}$ can refinement fail to halt, since at every such boundary point every measurement ball $B(x', \varepsilon)$ intersects one of the open decision regions. Then, why is it reasonable to restrict our attention to finite-precision tests? The first thing to notice is that this requirement is already more realistic than the usual one in which we impose no conditions (besides measurability) on the complexity of the regions. Furthermore, positive results are stronger: if it is informative to learn that a hypothesis is testable with arbitrary (measurable) regions, it is even more informative to learn that it is testable with arbitrary but finite precision. Negative results are still informative: if you cannot test a hypothesis with arbitrary but finite precision, then you certainly cannot test it with bounded precision.

For a generic FP-test, the termination of the precision-refinement protocol may depend on x and on n , with no a priori bound across the sample space. Real measurement devices, however, have a fixed bounded resolution. The stronger condition, in which a single fixed precision works at every sample size, is captured by the following definition.

Definition 2. A *bounded-precision test (BP-test)* $(\varphi_n)_{n \in \mathbb{N}}$ is a finite-precision test such that for some $\delta > 0$ we have $d_\infty(\{\varphi_n = 0\}, \{\varphi_n = 1\}) \geq \delta$ for all $n \in \mathbb{N}$.²

For a BP-test, the empiricist may fix a precision $\varepsilon < \delta/2$, since then $B(x', \varepsilon)$ has diameter at most $2\varepsilon < \delta$ and therefore cannot simultaneously intersect $\{\varphi_n = 0\}$ and $\{\varphi_n = 1\}$. The only residual ambiguity is between a single decision region $\{\varphi_n = i\}$ ($i \in \{0, 1\}$) and the suspension region $\{\varphi_n = 2\}$. Under uniform convergence $\sup_{\mathbb{P} \in H_0 \cup H_1} \mathbb{P}^n(\varphi_n = 2) \rightarrow 0$ (c.f. Theorem 6), the empiricist's asymptotic guarantees are unaffected by how this residual ambiguity is resolved.

Every FP-test (and hence also every BP-test) can be transformed into a binary test by merging $\{\varphi_n = 2\}$ with $\{\varphi_n = 0\}$ or $\{\varphi_n = 1\}$. Pointwise and uniform consistency are maintained by transforming it into a binary test in either way, while for maintaining one-sided error control one should be more careful. More details are given in Section 3.

The finite-precision measurement framework above is intrinsically topological, and instead of metric balls, one can consider other topological bases whose elements ought to correspond to finite-precision measurements. For more on topological and computational models of measurement, see Moore (1966); Vickers (1990); Kelly (1996); Beggs et al. (2010); Genin and Kelly (2017); Resende (2021).

1.2 Related literature

In a sufficiently regular parametric setting, Berger and Wald (1949) provide necessary and sufficient conditions for the existence of a test with Type-I and Type-II error control with $\alpha = \beta = 1/2$. A

¹Throughout the paper, we assume that the data comes from a distribution in H_0 or H_1 . Some of our mathematical results are more generally applicable to three-arm testing problems where the true $\mathbb{P}$ comes from H_0, H_1 or H_2 . However, in such cases one cannot verify based on finite-precision measurements whether a sample lands in $\{\varphi_n = 2\}$, and one should take H_2 into account in the definitions of consistency and uniform error control (see Section 2). In that case, the output $\{\varphi_n = 2\}$ is not restricted to the interpretation ‘suspension of judgement’.

²Here, d_∞ denotes metric on $\mathcal{X}^n$, taken to be the ℓ_∞ product of the metric on $\mathcal{X}$.

rather technical characterisation for the existence of uniformly consistent tests has been given by Berger (1951). A topological characterisation has been given by LeCam and Schwartz (1960) in terms of a topology of setwise convergence on the space $\cup_{n=1}^{\infty} \mathcal{P}(\mathcal{X})^n$ of all n -fold products of probability measures on $\mathcal{X}$. The downside of this condition is that it is hard to verify since this topology is not metrisable, not first countable and hence “not very easily accessible” (LeCam and Schwartz, 1960) — see also Kleijn (2022). In contrast, later work (and this paper as well) considers topological properties of H_0, H_1 as subsets of $\mathcal{P}(\mathcal{X})$, instead of the space of all n -fold products. Cover (1973) considered testing whether the bias of a coin is rational or irrational (example (1)), and found that there exists a consistent test for testing the null of rational bias versus the alternative of irrational bias, if a subset of Lebesgue measure zero is removed from the alternative. Dembo and Peres (1994) show that if H_0 and H_1 are disjoint sets which are F_σ in the weak topology on $W = H_0 \cup H_1$, then there exists a strongly consistent test. For the converse direction, they note that the hypotheses $H_0 := \{\delta_x : x \in [0, 1] \cap \mathbb{Q}\}$ and $H_1 := \{\delta_x : x \in [0, 1] \setminus \mathbb{Q}\}$ are discernible by the test $\varphi_n(x_1, \dots, x_n) := \mathbb{1}_{[0, 1] \setminus \mathbb{Q}}(x_1)$, but H_1 is not F_σ ³ so *some* regularity condition has to be imposed for such a topological characterisation of discernibility. Dembo and Peres (1994) prove this characterisation under the assumption that every measure in W has a $p > 1$ -integrable density, and Kleijn (2022) (Corollary 9.4.23) weakens this assumption to uniformly integrable densities. We show without any assumptions on the hypotheses H_0 and H_1 that consistent FP-testability is characterised by hypotheses that are F_σ in the weak topology, so requiring the regions $\{\varphi_n = 0\}$ and $\{\varphi_n = 1\}$ to be open is *the* regularity condition which resolves the issue raised by Dembo and Peres (1994). Ermakov (2017) provides characterisation theorems similar to our Theorems 3, 4 and 6, but for binary tests without regularity conditions on the critical regions, in terms of the topology of setwise convergence. For a comprehensive overview of the literature on consistent hypothesis testing, we refer to Kleijn (2022), Chapter 9.

Larsson et al. (2026) show that a test — as general $[0, 1]$ -valued random variable on the underlying measurable space, so without i.i.d. assumption — that satisfies the condition $\sup_{\mathbb{P} \in H_0} \mathbb{E}_{\mathbb{P}}[\varphi] < \inf_{\mathbb{P} \in H_1} \mathbb{E}_{\mathbb{P}}[\varphi]$ exists if and only if the weak* closures (in the space of bounded finitely additive measures) of the convex hulls of H_0 and H_1 are disjoint.

Our results in Section 3 are largely based on the work by Genin and Kelly (2017), who work under the assumption that the sample space $\mathcal{X}$ has a subbasis $\mathcal{O}$ such that $\mathbb{P}(\partial A) = 0$ for all $\mathbb{P} \in W$ and $A \in \mathcal{O}$ — this assumption is satisfied if e.g., all measures in W have a density with respect to some dominating measure (Bogachev, 2007, Proposition 8.2.8). The results of Genin and Kelly (2017) are applied to the study of causal discovery in Genin and Mayo-Wilson (2020); Genin (2021); Genin and Mayo-Wilson (2024).

Regarding conditional independence testing, Shah and Peters (2020) and Neykov et al. (2021) show, in the setting where the variables are real-valued and the probability measures have densities, that no conditional independence test with Type-I error control and consistency under the alternative exists. Lundborg et al. (2022) generalise this to a specific setting where the samples are $L^2([0, 1], \mathbb{R})$ functions, e.g. continuous-time stochastic processes. Györfi and Walk (2012) provide a conditional independence test and prove that it is strongly consistent, but Neykov et al. (2021) point out a mistake in their proof, so it remains an open question whether conditional independence is consistently testable. We answer this question for FP-testability by showing that if X, Y, Z take values in arbitrary Polish spaces $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ where $\mathcal{Z}$ has no isolated points, there exists no consistent FP-test for conditional independence $X \perp\!\!\!\perp Y \mid Z$.

In Section 5 we show that conditional independence is weakly closed under similar conditions as considered by Barbie and Gupta (2014); we provide a more direct, alternative proof. See Section 5.3 for further discussion of related literature. In Section 6, we provide sufficient conditions for the consistent testability of conditional independence with and without uniform error control under H_0 . Our conditions for testability with error control are similar to those considered by Warren (2021) and Neykov et al. (2021). To the best of our knowledge, our conditions for consistent testability are novel:

³This holds since $[0, 1] \setminus \mathbb{Q}$ is not F_σ and $x \mapsto \delta_x$ is a homeomorphism (Bogachev, 2007, Lemma 8.9.2).

we require that only one of the maps

$$\begin{aligned} z &\mapsto \mathbb{P}(X \mid Z = z), \\ z &\mapsto \mathbb{P}(Y \mid Z = z) \end{aligned}$$

satisfies a regularity condition. For a more in-depth comparison with existing literature, see Section 6.1.

1.3 The weak topology

We will characterise FP-testability in terms of topological properties of the hypotheses $H_0, H_1 \subseteq \mathcal{P}(\mathcal{X})$, where $\mathcal{X}$ is a separable metric space, and $\mathcal{P}(\mathcal{X})$ denotes the set of probability measures on the Borel σ -algebra $\mathcal{B}(\mathcal{X})$ on $\mathcal{X}$. Recall that a sequence of probability measures $\mathbb{P}_1(X), \mathbb{P}_2(X), \dots$ converges weakly to another probability measure $\mathbb{P}(X)$, denoted with $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$, if $\int f(x) d\mathbb{P}_n(x) \rightarrow \int f(x) d\mathbb{P}(x)$ for all bounded continuous functions $f : \mathcal{X} \rightarrow \mathbb{R}$. The *weak topology* on the space of Borel probability measures $\mathcal{P}(\mathcal{X})$ is the smallest topology that makes the maps $\mathbb{P} \mapsto \int f(x) d\mathbb{P}$ continuous for all bounded continuous f . Because $\mathcal{X}$ is separable, the weak topology is separable and metrisable, for example by the *bounded Lipschitz metric*⁴ d_{BL} , defined by

$$d_{BL}(\mathbb{P}_0(X), \mathbb{P}_1(X)) := \sup \left\{ \left| \int f d(\mathbb{P}_0 - \mathbb{P}_1) \right| : f \in \text{BL}(\mathcal{X}; \mathbb{R}) \right\},$$

where $\text{BL}(\mathcal{X}; \mathbb{R}) := \left\{ f : \mathcal{X} \rightarrow \mathbb{R} \mid \sup_{x \neq x'} \frac{|f(x) - f(x')|}{d(x, x')} \leq 1 \text{ and } \|f\|_\infty \leq 1 \right\}$ (Bogachev, 2007, Theorem 8.3.2). If $\mathcal{X}$ is complete, then d_{BL} is complete as well (Bogachev, 2007, Theorem 8.10.43). Since the weak topology is sequential, convergence in the weak topology coincides with weak convergence. Weak convergence $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$ is equivalent to the condition that $\liminf_n \mathbb{P}_n(A) \geq \mathbb{P}(A)$ for all $A \subseteq \mathcal{X}$ open, and to the condition that $\limsup_n \mathbb{P}_n(B) \leq \mathbb{P}(B)$ for all $B \subseteq \mathcal{X}$ closed. This is also known as the ‘Portmanteau theorem’ (Bogachev, 2007, Corollary 8.2.4(a)). A subbasis for the weak topology is given by sets $\{\mathbb{P} : \mathbb{P}(A) > q\}$, with $A \subseteq \mathcal{X}$ open and $q \in [0, 1]$; see for example Bogachev (2007), Section 8.2. This means that for any weakly open set $H \subseteq \mathcal{P}(\mathcal{X})$, there exist open sets $A_{ij} \subseteq \mathcal{X}$ and $q_{ij} \in [0, 1]$ such that $H = \bigcup_{i \in I} \bigcap_{j=1}^{m_i} \{\mathbb{P} : \mathbb{P}(A_{ij}) > q_{ij}\}$ for some index set I . Because $\mathcal{X}$ and hence $\mathcal{P}(\mathcal{X})$ are assumed to be separable, the index set I can be taken to be countable.

1.4 Convergence of empirical measures

For an i.i.d. sample $X_1, \dots, X_n \sim \mathbb{P}$, the *empirical measure* is $\mathbb{P}_n^X := \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$. By Varadarajan’s theorem (Dudley, 2002, Theorem 11.4.1) we have $\mathbb{P}_n^X \xrightarrow{w} \mathbb{P}$ almost surely; equivalently $d_{BL}(\mathbb{P}_n^X, \mathbb{P}) \rightarrow 0$ almost surely, and by dominated convergence the convergence also holds in expectation.

For Theorem 6 below we require the convergence to hold uniformly over a class of distributions: a set $W \subseteq \mathcal{P}(\mathcal{X})$ satisfies the *uniform Glivenko–Cantelli property* if

$$\lim_{n \rightarrow \infty} \sup_{\mathbb{P} \in W} \mathbb{E}_{\mathbb{P}}[d_{BL}(\mathbb{P}_n^X, \mathbb{P})] = 0. \quad (7)$$

This property is satisfied whenever W is uniformly tight (and so in particular whenever W is precompact in the weak topology, by Prokhorov’s theorem), but also for some non-tight families such as Gaussian location-scale families with bounded variance.

2 Preliminaries on consistent hypothesis testing and error control

First, we provide various definitions of consistency and error control. Although these notions themselves are not equivalent, it turns out that various equivalences can be derived for the *existence* of a

⁴The bounded Lipschitz metric is also known as the *Kantorovic–Rubinstein metric*. It is weaker than the total variation metric d_{TV} and the p -*Wasserstein metric* W_p : we have $d_{BL} \leq W_1 \leq W_p$, and strong equivalence of d_{BL} and W_1 if the measures under consideration have bounded support (Bogachev, 2007, Theorem 8.10.45).

(FP-)test with these properties. The definitions and results in this section apply to both FP-tests $\varphi_n : \mathcal{X}^n \rightarrow \{0, 1, 2\}$ (with $\{\varphi_n = 0\}$ and $\{\varphi_n = 1\}$ open) and binary tests $\varphi_n : \mathcal{X}^n \rightarrow \{0, 1\}$ without any conditions on the critical regions except measurability.

Definition 3. Let $(\varphi_n)_{n \in \mathbb{N}}$ be a (FP-)testing sequence for hypotheses $H_0, H_1 \subseteq \mathcal{P}(\mathcal{X})$. For $\mathbb{P} \in H_i$ ($i \in \{0, 1\}$) and $n \in \mathbb{N}$, define the *error processes*:⁵

$$\begin{aligned} e_n^w(\varphi, \mathbb{P}) &:= \mathbb{P}^n(\varphi_n \neq i), & (\text{weak}) \\ e_n^s(\varphi, \mathbb{P}) &:= \mathbb{P}^\infty(\exists m \geq n : \varphi_m \neq i), & (\text{strong}) \\ e_n^\Sigma(\varphi, \mathbb{P}) &:= \sum_{m=n}^{\infty} \mathbb{P}^m(\varphi_m \neq i). & (\text{summable}) \end{aligned}$$

For $(\star_0, \star_1) \in \{(\text{weak}, w), (\text{strong}, s), (\text{summable}, \Sigma)\}$ and $i \in \{0, 1\}$, we define the following properties:

- (φ_n) is $\star_0$ -consistent under H_i if

$$\lim_{n \rightarrow \infty} e_n^{\star_1}(\varphi, \mathbb{P}) = 0;$$

- (φ_n) is *uniformly* $\star_0$ -consistent under H_i if

$$\lim_{n \rightarrow \infty} \sup_{\mathbb{P} \in H_i} e_n^{\star_1}(\varphi, \mathbb{P}) = 0;$$

- for given $\alpha > 0$, (φ_n) has *asymptotic* $\star_0$ -level α under H_i if

$$\limsup_{n \rightarrow \infty} \sup_{\mathbb{P} \in H_i} e_n^{\star_1}(\varphi, \mathbb{P}) \leq \alpha;$$

- for given $\alpha > 0$, (φ_n) has $\star_0$ -level α under H_i if

$$\sup_{\mathbb{P} \in H_i} e_n^{\star_1}(\varphi, \mathbb{P}) \leq \alpha \quad \text{for all } n \in \mathbb{N}.$$

These notions are also depicted in Table 1. The error processes satisfy for all $\mathbb{P}$ and n

$$e_n^w(\varphi, \mathbb{P}) \leq e_n^s(\varphi, \mathbb{P}) \leq e_n^\Sigma(\varphi, \mathbb{P}), \quad (8)$$

where the first inequality follows from $\{\varphi_n \neq i\} \subseteq \{\exists m \geq n : \varphi_m \neq i\}$ and the second from the union bound. In particular, $\lim_n e_n^\Sigma(\varphi, \mathbb{P}) = 0$ (or equivalently $e_1^\Sigma(\varphi, \mathbb{P}) < \infty$) implies strong consistency by the Borel–Cantelli lemma. Since $e_n^s(\varphi, \mathbb{P})$ and $e_n^\Sigma(\varphi, \mathbb{P})$ are non-increasing in n , s -level α simplifies to $\sup_{\mathbb{P}} e_1^s(\varphi, \mathbb{P}) \leq \alpha$, and Σ -level α simplifies to $\sup_{\mathbb{P}} e_1^\Sigma(\varphi, \mathbb{P}) \leq \alpha$.

A testing sequence $(\varphi_n)_{n \in \mathbb{N}}$ is called *weakly (strongly) consistent* if it is weakly (strongly) consistent under both H_0 and H_1 . Hypotheses for which a consistent (FP-)test exists are sometimes referred to as *discernible* (Dembo and Peres, 1994). The following lemma gives a useful alternative characterisation of strong consistency.

Lemma 1. *A testing sequence (φ_n) is strongly consistent if and only if for all $i \in \{0, 1\}$ and all $\mathbb{P} \in H_i$,*

$$\mathbb{P}^\infty(\varphi_n \neq i \text{ for finitely many } n) = 1,$$

or equivalently, $\mathbb{P}^\infty(\liminf_n \{\varphi_n = i\}) = 1$.

Proof. The events $A_n := \{\exists m \geq n : \varphi_m \neq i\}$ are decreasing and we have $\bigcap_n A_n = \limsup_n \{\varphi_n \neq i\}$. By continuity of measures from above, $\mathbb{P}^\infty(\limsup_n \{\varphi_n \neq i\}) = \lim_{n \rightarrow \infty} \mathbb{P}^\infty(A_n) = \lim_{n \rightarrow \infty} e_n^s(\varphi, \mathbb{P})$. ■

The first non-trivial observation is that the existence of weakly and strongly consistent (FP-)tests are equivalent. For binary tests this was shown by Nobel (2006); we adapt his proof to FP-tests.

⁵The event $\{\varphi_n \neq i\}$ for $\mathbb{P} \in H_i$ encompasses both the wrong decision $\{\varphi_n = 1 - i\}$ and the suspension region $\{\varphi_n = 2\}$. Strictly speaking, the suspension region is not an error but a refusal to commit to a verdict; we nevertheless refer to the processes $e_n^\star$ as “error processes” for ease of exposition.

Table 1: Properties of a testing sequence (φ_n) under H_i ($i \in \{0, 1\}$), for given $\alpha > 0$.

	$\star = \text{weak}$	$\star = \text{strong}$	$\star = \text{summable}$
$\star$-consistent	$\lim_n e_n^w(\varphi, \mathbb{P}) = 0$	$\lim_n e_n^s(\varphi, \mathbb{P}) = 0$	$\lim_n e_n^\Sigma(\varphi, \mathbb{P}) = 0$
Unif. $\star$-consistent	$\limsup_n \sup_{\mathbb{P} \in H_i} e_n^w(\varphi, \mathbb{P}) = 0$	$\limsup_n \sup_{\mathbb{P} \in H_i} e_n^s(\varphi, \mathbb{P}) = 0$	$\limsup_n \sup_{\mathbb{P} \in H_i} e_n^\Sigma(\varphi, \mathbb{P}) = 0$
Asymp. $\star$-level α	$\limsup_n \sup_{\mathbb{P} \in H_i} e_n^w(\varphi, \mathbb{P}) \leq \alpha$	$\limsup_n \sup_{\mathbb{P} \in H_i} e_n^s(\varphi, \mathbb{P}) \leq \alpha$	$\limsup_n \sup_{\mathbb{P} \in H_i} e_n^\Sigma(\varphi, \mathbb{P}) \leq \alpha$
$\star$-level α	$\sup_{\mathbb{P} \in H_i} e_n^w(\varphi, \mathbb{P}) \leq \alpha \quad \forall n$	$\sup_{\mathbb{P} \in H_i} e_n^s(\varphi, \mathbb{P}) \leq \alpha$	$\sup_{\mathbb{P} \in H_i} e_n^\Sigma(\varphi, \mathbb{P}) \leq \alpha$

Theorem 1. *Given a pair of hypotheses $H_0, H_1 \subseteq \mathcal{P}(\mathcal{X})$, there exists a weakly consistent (FP-)test if and only if there exists a strongly consistent (FP-)test.*

Proof. Any strongly consistent test is also weakly consistent. Conversely, let $\varphi_n : \mathcal{X}^n \rightarrow \{0, 1, 2\}$ be weakly consistent, let $k_n := \lfloor \log(n) \rfloor$ and $m_n := \lfloor n/k_n \rfloor$. For $i \in \{0, 1\}$ define the random variable $Y_{n,j}^i := \mathbb{1}\{\varphi_{k_n}(X_{k_n j+1}, \dots, X_{k_n j+k_n}) = i\}$ such that $Y_{n,0}^i, \dots, Y_{n,m_n-1}^i$ are i.i.d., and let

$$\psi_n(X_1, \dots, X_n) := \begin{cases} 0 & \text{if } \frac{1}{m_n} \sum_{j=0}^{m_n-1} Y_{n,j}^0 > 1/2; \\ 1 & \text{if } \frac{1}{m_n} \sum_{j=0}^{m_n-1} Y_{n,j}^1 > 1/2; \\ 2 & \text{otherwise.} \end{cases}$$

(In the case that φ_n is a binary test, as corresponding binary test ψ_n is obtained by replacing one of the strict inequalities with a non-strict inequality.) Fix $i \in \{0, 1\}$. For $\mathbb{P} \in H_i$ we have $\mu_n^i := \mathbb{E}_{\mathbb{P}}[Y_{n,j}^i] = \mathbb{P}^n(\varphi_{k_n}(X_{k_n j+1}, \dots, X_{k_n j+k_n}) = i) \rightarrow 1$ by weak consistency of φ_n , and hence for every $\varepsilon \in (0, 1/2)$ there is an $N_{\mathbb{P}}$ such that $1/2 \leq \mu_n^i - \varepsilon$ for all $n \geq N_{\mathbb{P}}$, so using Hoeffding's inequality this gives

$$\begin{aligned} \mathbb{P}^n(\psi_n \neq i) &= \mathbb{P}^n\left(\frac{1}{m_n} \sum_{j=0}^{m_n-1} Y_{n,j}^i \leq 1/2\right) \leq \mathbb{P}^n\left(\frac{1}{m_n} \sum_{j=0}^{m_n-1} (Y_{n,j}^i - \mu_n^i) \leq -\varepsilon\right) \\ &\leq \mathbb{P}^n\left(\left|\frac{1}{m_n} \sum_{j=0}^{m_n-1} (Y_{n,j}^i - \mu_n^i)\right| > \varepsilon\right) \leq 2e^{-2m_n\varepsilon^2}, \end{aligned}$$

we get $\sum_{n \geq N_{\mathbb{P}}} e^{-2m_n\varepsilon^2} < \infty$, hence $\sum_{n=1}^{\infty} \mathbb{P}^n(\psi_n \neq i) < \infty$, so the result follows from the Borel-Cantelli lemma. $\blacksquare$

A second equivalence concerns the various notions of uniform error control in Definition 3. Similar as for pointwise consistency, a testing sequence $(\varphi_n)_{n \in \mathbb{N}}$ is called *uniformly weakly (strongly) consistent* if it is uniformly weakly (strongly) consistent under both H_0 and H_1 . For binary tests, the equivalence of uniform weak consistency and uniform strong consistency was shown by Pfanzagl (1968).

Theorem 2. *Let a pair of hypotheses $H_0, H_1 \subseteq \mathcal{P}(\mathcal{X})$ be given, and let $K \subseteq \{H_0, H_1\}$ be a set of hypotheses. The following are equivalent, for any $\star, \star' \in \{w, s, \Sigma\}$:*

- (a) *for every $\alpha > 0$, there exists a consistent (FP-)testing sequence (φ_n) with asymptotic $\star$ -level α under H_i for all $H_i \in K$:*

$$\limsup_{n \rightarrow \infty} \sup_{\mathbb{P} \in H_i} e_n^{\star}(\varphi, \mathbb{P}) \leq \alpha;$$

- (b) *there exists a consistent (FP-)testing sequence (φ_n) which is uniformly $\star'$ -consistent under H_i for all $H_i \in K$:*

$$\lim_{n \rightarrow \infty} \sup_{\mathbb{P} \in H_i} e_n^{\star'}(\varphi, \mathbb{P}) = 0.$$

If for only a single hypothesis the errors are controlled, say $K = \{H_i\}$, the above are also equivalent to the following, for any $\star'' \in \{w, s, \Sigma\}$:

(c) for every $\alpha > 0$ there exists a consistent (FP-)testing sequence (φ_n) with $\star''$ -level α under H_i :

$$\sup_{\mathbb{P} \in H_i} e_n^{\star''}(\varphi, \mathbb{P}) \leq \alpha \quad \text{for all } n \in \mathbb{N}.$$

Throughout, it is equivalent to consider weak or strong consistency of the tests.

Proof. The chain of inequalities (8) implies that each condition with $\star = \Sigma$ implies the same condition with $\star = s$, which implies the same condition with $\star = w$. Similarly, strong consistency implies weak consistency. Hence, in each equivalence below, it suffices to show that the weakest version (with $\star = w$ and weak consistency) implies the strongest (with $\star = \Sigma$ and strong consistency).

Equivalence of (a) and (b). The direction (b) $\implies$ (a) is immediate from (8). For the converse, we show that (a) with $\star = w$ and weak consistency implies (b) with $\star' = \Sigma$ and strong consistency. Let $\varepsilon \in (0, 1/2)$ and let (φ_n) be weakly consistent with $\sup_{\mathbb{P} \in H_i} \mathbb{P}^n(\varphi_n \neq i) \leq 1/2 - \varepsilon$ for all $n \geq N$ and all $H_i \in K$. With ψ_n as defined in the proof of Theorem 1, $N_{\mathbb{P}} = N$ is independent of $\mathbb{P}$, so $\mathbb{P}^n(\psi_n \neq i) \leq 2e^{-2m_n\varepsilon^2}$ uniformly over $\mathbb{P} \in H_i$ for all $n \geq N$. Hence $\sup_{\mathbb{P} \in H_i} \sum_{n \geq N} \mathbb{P}^n(\psi_n \neq i) \leq \sum_{n \geq N} 2e^{-2m_n\varepsilon^2} < \infty$, and as the tail of a convergent series, $\sup_{\mathbb{P} \in H_i} e_n^{\Sigma}(\psi, \mathbb{P}) \rightarrow 0$.

Equivalence with (c) when $K = \{H_i\}$. Without loss of generality, let $K = \{H_0\}$. The direction (c) $\implies$ (a) is immediate from (8). For the converse, we show that (b) with $\star' = \Sigma$ implies (c) with $\star'' = \Sigma$: for every $\alpha > 0$ there is an N such that $\sup_{\mathbb{P} \in H_0} e_n^{\Sigma}(\varphi, \mathbb{P}) \leq \alpha$ for all $n \geq N$. Then $\psi_n := \mathbb{1}\{n \geq N\}\varphi_n$ satisfies $\sup_{\mathbb{P} \in H_0} e_n^{\Sigma}(\psi, \mathbb{P}) \leq \alpha$ for all n . $\blacksquare$

By Theorem 2, all notions of uniform error control in Definition 3 collapse into a single concept.

Definition 4. We say that a pair of hypotheses H_0 and H_1 are *consistently FP-testable* if there exists a weakly (or equivalently strongly, by Theorem 1) consistent FP-test for them, and that they are *consistently FP-testable with uniform error control under H_i* if there exists a consistent FP-test that satisfies any of the equivalent conditions of Theorem 2.

3 Main results

Let $\mathcal{X}$ be a separable metric space, and let H_0, H_1 be disjoint sets of Borel probability measures on $\mathcal{X}$.

Theorem 3. *The following are equivalent:*

1. *there exists a consistent FP-test;*
2. *H_0 and H_1 are F_{σ} in the weak topology on $W := H_0 \cup H_1$.*

Theorem 4. *For $i \in \{0, 1\}$, the following are equivalent:*

1. *there exists a consistent FP-test φ_n with*

$$\lim_{n \rightarrow \infty} \sup_{\mathbb{P} \in H_i} \mathbb{P}^n(\varphi_n = 1 - i) = 0;$$

2. *there exists a consistent FP-test φ_n with uniform error control under H_i , e.g.*

$$\lim_{n \rightarrow \infty} \sup_{\mathbb{P} \in H_i} \mathbb{P}^n(\varphi_n \neq i) = 0;$$

3. *H_i is closed in the weak topology on $W := H_0 \cup H_1$;*

Theorem 5. *The following are equivalent:*

1. there exists a consistent FP-test φ_n with for all $i \in \{0, 1\}$:

$$\lim_{n \rightarrow \infty} \sup_{\mathbb{P} \in H_i} \mathbb{P}^n(\varphi_n = 1 - i) = 0;$$

2. H_0 and H_1 are clopen in the weak topology on $W := H_0 \cup H_1$.

The next theorem characterises the strongest mode of testability: uniform consistency with bounded measurement precision. Under a precompactness assumption, this is equivalent to metric separation of H_0 and H_1 in the bounded Lipschitz metric.

Theorem 6. *For the following statements we have the implications $1 \implies 2 \implies 3$:*

1. $d_{BL}(H_0, H_1) > 0$ and $W := H_0 \cup H_1$ satisfies the uniform Glivenko–Cantelli property;

2. there exists a uniformly consistent BP-test, e.g. with for all $i \in \{0, 1\}$

$$\lim_{n \rightarrow \infty} \sup_{\mathbb{P} \in H_i} \mathbb{P}^n(\varphi_n \neq i) = 0;$$

3. H_0 and H_1 have disjoint closures in $\mathcal{P}(\mathcal{X})$.

If additionally $\mathcal{X}$ is complete and W is precompact in the weak topology, then it satisfies the uniform Glivenko–Cantelli property and $1 \iff 2 \iff 3$.

The proofs are given below in Section 3.1. Throughout one can equivalently consider strong consistency and weak consistency, by Theorem 1. Theorem 2 shows that the uniform error control of Theorem 4, which is now stated as uniform consistency under H_i , can equivalently be stated as controlling the level under H_i , either asymptotically, or for finite samples. The uniform consistency of Theorem 6 can equivalently be stated as control of the asymptotic level under H_0 and H_1 . Similarly, it can be shown that the ‘uniform consistency’ of Theorem 5 can equivalently be stated as asymptotic or finite-sample error control of these specific kinds of errors.

Theorem 4 shows that uniform control of $\mathbb{P}^n(\varphi_n \neq i)$ is equivalent to uniform control of $\mathbb{P}^n(\varphi_n = 1 - i)$. For two-sided error control, the distinction matters: pointwise convergence of the suspension region (Theorem 5) corresponds to different topological properties of the hypotheses than uniform convergence of the suspension region (Theorem 6).

Theorem 6 shows that disjoint closures of the hypotheses is necessary for the existence of a uniformly consistent BP-test. The sufficient conditions are metric separation $d_{BL}(H_0, H_1) > 0$ and the uniform Glivenko–Cantelli property (7). Note that disjoint closures do not imply metric separation without precompactness: for instance, $H_0 = \{\delta_k : k \in \mathbb{N}\}$ and $H_1 = \{\delta_{k+1/k} : k \in \mathbb{N}\}$ on $\mathbb{R}$ have disjoint closures but $d_{BL}(H_0, H_1) = 0$. If $H_0 \cup H_1$ is precompact (equivalently, uniformly tight by Prokhorov’s theorem), disjoint closures implies metric separation and (7) holds automatically, so the two directions yield an equivalence. The implication $1 \implies 2$ also covers non-precompact examples: for instance, one can uniformly consistently test whether the mean of a Gaussian with bounded variance is smaller than -1 or larger than 1 , since $d_{BL}(H_0, H_1) > 0$ and the uniform Glivenko–Cantelli property holds.

The topological condition of each theorem implies the topological condition of the previous theorem: metric separation implies the hypotheses to be clopen in the subspace topology, which implies H_0 to be closed, which implies H_0 and H_1 to be F_σ (since the weak topology is a separable metric topology).

The four theorems yield FP-tests of differing precision strength. The constructions in the proofs of Theorems 3, 4 and 5 produce critical regions that are open and disjoint in $\mathcal{X}^n$, so the verdict can be determined by measurements of (unbounded) finite precision — the precision required may depend on the location of the sample within the critical region (see Section 1.1). The construction in the proof of Theorem 6 is stronger: it is a BP-test with metric separation of the critical regions by $0 < \delta < d_{BL}(H_0, H_1)$ that can be taken arbitrarily close to $d_{BL}(H_0, H_1)$.

These results are readily applicable to the example from the introduction, since the parameter space $[0, 1]$ with the Euclidean topology is homeomorphic to the set of Bernoulli distributions equipped with the weak topology. The hypothesis H_1 in (1) is not F_σ , so there does not exist a consistent FP-test

for this problem. The hypotheses from (2) are both F_σ so there exists a consistent test, but no form of uniform error control is possible since neither H_0 nor H_1 is closed. In (3) H_0 is closed, so there exists a consistent FP-test with uniform error control under H_0 . Both hypotheses of (4) are clopen in the relative topology on W , so there exists a consistent FP-test with the type of error control as in Theorem 5, but uniform consistency as in Theorem 6 is not feasible. In (5) the two hypotheses are relatively compact and have disjoint closures in the ambient space, so there exists a uniformly consistent test.

In general, these sufficient conditions for various types of testability are not constructive: the tests of Theorems 3, 4 and 5 are constructed by expressing the hypotheses in terms of the subbasis elements $\{\mathbb{P} \in W : \mathbb{P}(A) > q\}$ of the weak topology. For example, in Theorem 4 one might know that H_1 is open without knowing an explicit representation in terms of the subbasis elements. However, for the examples (2), (3) and (4) these subbasis representations are easily derived, allowing us to give explicit tests, following the constructions in the proofs. We explicitly provide the test for the pair of hypotheses $H_0 = [0, 1] \cap \mathbb{Q}$ and $H_1 = [0, 1] \cap (\mathbb{Q} + \sqrt{2})$ of (4) in Example 1 below. The test of Theorem 6 requires the computation of the BL-distance between the empirical measure and the hypotheses, which might be computationally intractable as well.

These sufficient conditions for FP-testability also imply binary testability with corresponding modes of error control, by joining the suspension region $\{\varphi_n = 2\}$ with either $\{\varphi_n = 0\}$ or $\{\varphi_n = 1\}$. In Theorem 3, merging the suspension region with $\{\varphi_n = 0\}$ or $\{\varphi_n = 1\}$ both give a consistent test. Actually, the topological condition of H_0 and H_1 being F_σ is equivalent to the existence of two binary testing sequences: one with $\{\varphi_n = 0\}$ open, and one with $\{\varphi_n = 1\}$ open. This can rather straightforwardly be deduced from the proof of Theorem 3. For the test of clause 2 of Theorem 4 it does not matter for maintaining error control whether $\{\varphi_n = 2\}$ is joined with $\{\varphi_n = 0\}$ or $\{\varphi_n = 1\}$, but for the test of clause 1 the suspension region must be joined with $\{\varphi_n = 0\}$ to maintain the error control under H_0 . It can be shown that H_0 being closed is equivalent to the existence of a binary testing sequence with $\{\varphi_n = 1\}$ open and uniform error control under H_0 . For Theorem 5 the FP-test cannot be converted to a binary test while maintaining the same error control. Finally, for Theorem 6 the binary test can be constructed both ways while maintaining uniform consistency.

Results similar to Theorems 3 – 6 exist in the literature. Theorem 3 generalises Dembo and Peres (1994) (Theorem 2) who provide the similar characterisation that, under the assumption that all measures in W have uniformly integrable densities, there exists a consistent binary test if and only if the hypotheses are F_σ in the weak topology. Genin and Kelly (2017) (Theorem 4.3) drop the uniform integrability assumption (but the assumption of having densities remains), for which they show that the F_σ condition is equivalent to the existence of a binary testing sequence $(\varphi_n)_{n \in \mathbb{N}}$ with $\mathbb{P}(\partial\{\varphi_n = 0\}) = \mathbb{P}(\partial\{\varphi_n = 1\}) = 0$ for all $\mathbb{P} \in W$. Ermakov (2017) (Theorem 4.4) considers the topology of setwise convergence instead of the weak topology for which he shows – under the assumption that W is contained in a σ -compact set – that the F_σ condition is equivalent to the existence of a binary testing sequence, without any regularity of the critical regions.

An analogue of Theorem 4 is also shown by Genin and Kelly (2017) (Theorem 4.1) under the assumption of having densities, for binary tests with $\mathbb{P}(\partial\{\varphi_n = 0\}) = \mathbb{P}(\partial\{\varphi_n = 1\}) = 0$ for all $\mathbb{P} \in W$. Ermakov (2017) (Theorem 4.3) shows that when H_0 and H_1 are contained in respectively a compact set and a σ -compact set, consistent binary testability with uniform consistency under H_0 is equivalent to H_0 being closed and H_1 being F_σ in the topology of setwise convergence.

Theorem 5 is also given by Genin (2018) (Theorem 3.2.3) under the assumption of having densities, for binary tests with $\mathbb{P}(\partial\{\varphi_n = 0\}) = \mathbb{P}(\partial\{\varphi_n = 1\}) = 0$ for all $\mathbb{P} \in W$.

Theorem 6 is similar to Ermakov (2017) (Theorem 4.1), who shows that if W is contained in a compact set in the topology of setwise convergence, then H_0 and H_1 having disjoint closures is equivalent to the existence of a consistent binary test. Note that compactness in the topology of setwise convergence implies that the weak topology and the topology of setwise convergence coincide, so our results give weaker sufficient conditions for the existence of uniformly consistent tests.

Example 1. Denoting $\mathbb{P}_p = \text{Bernoulli}(p)$, the hypotheses $H_0 : p \in [0, 1] \cap \mathbb{Q}$ and $H_1 : p \in [0, 1] \cap (\mathbb{Q} + \sqrt{2})$ can be specified as the countable unions of singletons $H_0 = \{\mathbb{P}_{p_m^0} : m \in \mathbb{N}\}$ and $H_1 = \{\mathbb{P}_{p_m^1} : m \in \mathbb{N}\}$, where p_m^0 and p_m^1 are enumerations of $[0, 1] \cap \mathbb{Q}$ and $[0, 1] \cap (\mathbb{Q} + \sqrt{2})$ respectively. For testing the

closed set $\{\mathbb{P}_{p_m^i}\}$ against $W \setminus \{\mathbb{P}_{p_m^i}\}$, Theorem 4 gives the test

$$\varphi_{i,n}^m(X_1, \dots, X_n) = \begin{cases} i & \text{if } \bar{X}_n \in p_m^i + I_n; \\ 1 - i & \text{if } \bar{X}_n \notin p_m^i + I_n, \end{cases}$$

where we write $p + I_n := (p - t_n^\alpha, p + t_n^\alpha)$ and $t_n^\alpha := \sqrt{\frac{1}{2n} \ln(\pi^2 n^2 / 6\alpha)}$. This gives us the strongly consistent test

$$\varphi_n(X_1, \dots, X_n) = \begin{cases} 0 & \text{if } \exists m \leq n : \bar{X}_n \in p_m^0 + I_n \text{ and } \bar{X}_n \notin p_k^1 + I_n \text{ for all } k \leq m; \\ 1 & \text{if } \exists m \leq n : \bar{X}_n \in p_m^1 + I_n \text{ and } \bar{X}_n \notin p_k^0 + I_n \text{ for all } k \leq m; \\ 2 & \text{otherwise.} \end{cases}$$

Another way to see that we indeed have strong consistency, is that if $X_1, X_2, \dots \sim \mathbb{P}_{p^*} \in H_1$, there is an m such that $p_m^1 = p^*$. For sufficiently large n we have $\bar{X}_n \in p^* + I_n$ and $\bar{X}_n \notin p_k^0 + I_n$ for all $k \leq m$.

3.1 Proofs

The proofs that the existence of certain tests imply certain topological conditions on the hypotheses are stand-alone. There is however a dependency between the proofs that certain topological conditions imply the existence of FP-tests with the desired properties. We first prove Theorem 4. From that one, we can build under F_σ conditions consistent tests in Theorem 3, and under clopen conditions we can build tests with the two-sided error control as required in Theorem 5. To prove Theorem 6, we use an entirely different proof strategy using uniform convergence of empirical measures.

3.1.1 Proof of Theorem 4

We prove the result for $i = 0$, corresponding to uniform error control under H_0 , which is considered to be weakly closed. The main argument for the existence of a test with uniform error control is that if H_1 is open, we can write it as a countable union of finite intersections of subbasis elements $\{\mathbb{P} \in W : \mathbb{P}(A) > q\}$. For each disjoint pair of subbasis elements there exists a test with the required error control. The proofs largely follow the structure of the proof of Genin and Kelly (2017), Theorem 4.1.

Lemma 2. *Let $W \subseteq \mathcal{P}(\mathcal{X})$ be given, let $A \subseteq \mathcal{X}$ be open and let $q \in [0, 1]$. For the hypotheses $H_0 := \{\mathbb{P} \in W : \mathbb{P}(A) \leq q\}$ and $H_1 = \{\mathbb{P} \in W : \mathbb{P}(A) > q\}$, for every $\alpha > 0$ there exists a strongly consistent FP-testing sequence $(\varphi_n)_{n \in \mathbb{N}}$ with $\sup_{\mathbb{P} \in H_0} \sum_{n=1}^{\infty} \mathbb{P}^n(\varphi_n \neq 0) \leq \alpha$.*

Proof. Let $\alpha > 0$ be given, let $A_{1/n}^c := \bigcup_{a \in A^c} B(a, 1/n)$ be the $1/n$ -neighborhood of A^c , and define

$$\varphi_n(X_1, \dots, X_n) := \begin{cases} 0 & \text{if } \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{A_{1/n}^c}(X_i) > 1 - q - t_n; \\ 1 & \text{if } \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\text{ext}(A_{1/n}^c)}(X_i) \geq q + t_n; \\ 2 & \text{otherwise,} \end{cases}$$

where $t_n := \sqrt{\frac{1}{2n} \ln(\pi^2 n^2 / 6\alpha)}$.

For $\mathbb{P} \in H_0$ we have $\mathbb{P}(A) \leq q$, so the fact that $A_{1/n}^c \supseteq A^c$ and Hoeffding's inequality gives

$$\begin{aligned} \mathbb{P}^n(\varphi_n \neq 0) &= \mathbb{P}^n \left(\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{A_{1/n}^c}(X_i) \leq 1 - q - t_n \right) \leq \mathbb{P}^n \left(\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{A^c}(X_i) \leq 1 - q - t_n \right) \\ &= \mathbb{P}^n \left(\frac{1}{n} \sum_{i=1}^n \mathbb{1}_A(X_i) \geq q + t_n \right) \leq \mathbb{P}^n \left(\frac{1}{n} \sum_{i=1}^n \mathbb{1}_A(X_i) \geq \mathbb{P}(A) + t_n \right) \leq e^{-2nt_n^2} = \frac{6\alpha}{\pi^2 n^2}, \end{aligned}$$

hence $\sum_{n=1}^{\infty} \mathbb{P}^n(\varphi_n \neq 0) \leq \alpha$. Strong consistency under the null then follows from the Borel-Cantelli lemma.

If $\mathbb{P} \in H_1$ then $\mathbb{P}(A) > q$. Since $A = \bigcup_n \text{ext}(A_{1/n}^c)$, we have that $\mathbb{P}(\text{ext}(A_{1/n}^c)) \uparrow \mathbb{P}(A)$. Therefore, there is an $M \in \mathbb{N}$ such that $\mathbb{P}(\text{ext}(A_{1/M}^c)) > q$. Since $\text{ext}(A_{1/n}^c) \supseteq \text{ext}(A_{1/M}^c)$ for all $n \geq M$ we have for all these n that $\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\text{ext}(A_{1/n}^c)}(X_i) \geq \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\text{ext}(A_{1/M}^c)}(X_i)$, which by the strong law of large numbers converges almost surely to $\mathbb{P}(\text{ext}(A_{1/M}^c)) > q$. Combined with the fact that $t_n \downarrow 0$ we almost surely have that $\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\text{ext}(A_{1/n}^c)}(X_i) \geq q + t_n$ for all sufficiently large n , so $\varphi_n \rightarrow 1$ a.s.

To prove that $\{\varphi_n = 0\}$ is open, let for $\gamma \in \{0, 1\}^n$ the set $(A_{1/n}^c)^\gamma \subseteq \mathcal{X}^n$ be the Cartesian product of n sets, where the i -th set is $A_{1/n}^c$ if $\gamma_i = 1$, and $\mathcal{X}$ if $\gamma_i = 0$. For example, $\gamma = (1, \dots, 1, 0)$ gives $(A_{1/n}^c)^\gamma = A_{1/n}^c \times \dots \times A_{1/n}^c \times \mathcal{X}$. Since $A_{1/n}^c$ is open, so is $(A_{1/n}^c)^\gamma$, hence $\{\varphi_n = 0\} = \bigcup\{(A_{1/n}^c)^\gamma : \gamma \in \{0, 1\}^n, |\gamma| > n(1 - q - t_n)\}$ is open as well. Since $\text{ext}(A_{1/n}^c)$ is open, we have analogously that $\{\varphi_n = 1\}$ is open as well. $\blacksquare$

The following lemma shows that for pairs of hypotheses which are consistently testable with uniform error control under the null, the alternative hypotheses enjoy a topological structure: they are closed under countable unions and finite intersections. For convenience in proving Theorem 4, the result is stated in terms of strong consistency and Σ -level α (Definition 3), but by Theorem 2 also holds for weak consistency and w -level α .

Lemma 3. *Let $\{(H_0^{ij}, H_1^{ij}) : (i, j) \in \mathbb{N}^2\}$ be pairs of disjoint hypotheses such that for every pair (H_0^{ij}, H_1^{ij}) and every $\alpha^{ij} > 0$ there exists a strongly consistent FP-test φ_n^{ij} with $\sup_{\mathbb{P} \in H_0^{ij}} \sum_{n=1}^{\infty} \mathbb{P}^n(\varphi_n^{ij} \neq 0) \leq \alpha^{ij}$, then for the hypotheses $H_0 := \bigcap_{i=1}^{\infty} \bigcup_{j=1}^{m_i} H_0^{ij}$ and $H_1 := \bigcup_{i=1}^{\infty} \bigcap_{j=1}^{m_i} H_1^{ij}$ and any $\alpha > 0$ there exists a strongly consistent FP-test φ_n with $\sup_{\mathbb{P} \in H_0} \sum_{n=1}^{\infty} \mathbb{P}^n(\varphi_n \neq 0) \leq \alpha$.*

Proof. Let $\alpha > 0$ be given. For the hypotheses (H_0^{ij}, H_1^{ij}) , let φ_n^{ij} be a strongly consistent FP-test with $\sum_{n=1}^{\infty} \mathbb{P}^n(\varphi_n^{ij} \neq 0) \leq \alpha/2^i$. Let φ_n be the FP-test defined by

$$\begin{aligned} \{\varphi_n = 0\} &:= \bigcap_{i=1}^n \bigcup_{j=1}^{m_i} \{\varphi_n^{ij} = 0\} \\ \{\varphi_n = 1\} &:= \bigcup_{i=1}^n \bigcap_{j=1}^{m_i} \{\varphi_n^{ij} = 1\} \\ \{\varphi_n = 2\} &:= \mathcal{X}^n \setminus (\{\varphi_n = 0\} \cup \{\varphi_n = 1\}), \end{aligned}$$

then φ_n has uniform error control under the null: if $\mathbb{P} \in H_0$, then for every $i \in \mathbb{N}$ there is a $j_i \leq m_i$ such that $\mathbb{P} \in H_0^{ij_i}$, and since $\{\varphi_n \neq 0\} = \bigcup_{i=1}^n \bigcap_{j=1}^{m_i} \{\varphi_n^{ij} \neq 0\} \subseteq \bigcup_{i=1}^{\infty} \{\varphi_n^{ij_i} \neq 0\}$ we have

$$\sum_{n=1}^{\infty} \mathbb{P}^n(\varphi_n \neq 0) \leq \sum_{n=1}^{\infty} \mathbb{P}^n(\bigcup_{i=1}^{\infty} \{\varphi_n^{ij_i} \neq 0\}) \leq \sum_{n=1}^{\infty} \sum_{i=1}^{\infty} \mathbb{P}^n(\varphi_n^{ij_i} \neq 0) \leq \alpha.$$

The Borel-Cantelli lemma then gives strong consistency under H_0 .

For every $\mathbb{P} \in H_1$ there is an $i \in \mathbb{N}$ such that $\mathbb{P} \in \bigcap_{j=1}^{m_i} H_1^{ij}$. Because of the inclusion $\bigcap_{j=1}^{m_i} \{\varphi_n^{ij} = 1\} \subseteq \{\varphi_n = 1\}$ and the Fréchet inequality⁶ we have for $n \geq i$ that

$$\begin{aligned} \mathbb{P}^\infty(\liminf_n \{\varphi_n = 1\}) &= \mathbb{P}^\infty(\liminf_{n \geq i} \{\varphi_n = 1\}) \\ &\geq \mathbb{P}^\infty(\liminf_n \bigcap_{j=1}^{m_i} \{\varphi_n^{ij} = 1\}) \geq \sum_{j=1}^{m_i} \mathbb{P}^\infty(\liminf_n \{\varphi_n^{ij} = 1\}) - m_i + 1 = 1, \end{aligned}$$

so φ_n is strongly consistent. $\blacksquare$

We are now able to prove Theorem 4:

Proof.

⁶For measurable sets $A_1, \dots, A_n$ we have $\mathbb{P}(\bigcap_{i=1}^n A_i) \geq \sum_{i=1}^n \mathbb{P}(A_i) - n + 1$.

1 $\implies$ 3 By Theorem 2, 1 is equivalent to the existence (for every $\alpha > 0$) of a weakly consistent FP-test with $\sup_{\mathbb{P} \in H_0} \mathbb{P}^n(\varphi_n = 1) \leq \alpha$ for all n . If H_0 is not closed, then there is a $\mathbb{Q} \in H_1$ and a sequence $\{\mathbb{P}_m\}_{m \in \mathbb{N}}$ in H_0 such that $\mathbb{P}_m \xrightarrow{w} \mathbb{Q}$. The map $\mathbb{P} \mapsto \mathbb{P}^n$ is continuous (Billingsley, 1999, Theorem 2.8) hence $\mathbb{P}_m^n \xrightarrow{w} \mathbb{Q}^n$. Let φ_n be an FP-test with $\sup_{\mathbb{P} \in H_0} \mathbb{P}^n(\varphi_n = 1) \leq \alpha$ for all n , then $\{\varphi_n = 1\}$ is open so by the Portmanteau theorem we have $\mathbb{Q}^n(\varphi_n = 1) \leq \liminf_m \mathbb{P}_m^n(\varphi_n = 1) \leq \alpha$, and hence φ_n cannot be weakly consistent at $\mathbb{Q}$.

3 $\implies$ 2 If H_1 is open, then we can write H_0 and H_1 as

$$H_0 = \bigcap_{i=1}^{\infty} \bigcup_{j=1}^{m_i} \{\mathbb{P} \in W : \mathbb{P}(A_{ij}) \leq q_{ij}\} \quad H_1 = \bigcup_{i=1}^{\infty} \bigcap_{j=1}^{m_i} \{\mathbb{P} \in W : \mathbb{P}(A_{ij}) > q_{ij}\}$$

for $A_{ij} \subseteq \mathcal{X}$ open and $q_{ij} \in [0, 1]$. By Lemma 2, for each of these pairs of hypotheses $H_0^{ij} := \{\mathbb{P} \in W : \mathbb{P}(A_{ij}) \leq q_{ij}\}$ and $H_1^{ij} := \{\mathbb{P} \in W : \mathbb{P}(A_{ij}) > q_{ij}\}$, for each $\alpha^{ij} > 0$ there exists a strongly consistent FP-test φ_n^{ij} with $\sup_{\mathbb{P} \in H_0} \sum_{n=1}^{\infty} \mathbb{P}^n(\varphi_n^{ij} \neq 0) \leq \alpha^{ij}$, and by Lemma 3 this implies that for H_0, H_1 there exists a strongly consistent FP-test φ_n with $\sup_{\mathbb{P} \in H_0} \sum_{n=1}^{\infty} \mathbb{P}^n(\varphi_n \neq 0) < \alpha$. The result now follows from Theorem 2.

2 $\implies$ 1 Immediate. ■

3.1.2 Proof of Theorem 3

We now show that consistent FP-testability is equivalent to the hypotheses being F_σ in W .

Proof.

1 $\implies$ 2 Let φ_n be a weakly consistent FP-test. For $i \in \{0, 1\}$ we have $\mathbb{P} \in H_i$ if and only if there is an $m \in \mathbb{N}$ such that $\mathbb{P}^n(\varphi_n = i) > 2/3$ for all $n \geq m$, hence we can write

$$H_0 = \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} \{\mathbb{P} \in W : \mathbb{P}^n(\varphi_n = 1) \leq 1/3\}$$

$$H_1 = \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} \{\mathbb{P} \in W : \mathbb{P}^n(\varphi_n = 0) \leq 1/3\}.$$

Since sets of the form $\{\mathbb{P} \in W : \mathbb{P}(A) > q\}$ with $A \subseteq \mathcal{X}$ open are open in the weak topology on W , and both critical regions $\{\varphi_n = 0\}$ and $\{\varphi_n = 1\}$ are open, the set $\{\mathbb{P} \in W : \mathbb{P}^n(\varphi_n = i) \leq 1/3\}$ is closed, so H_0 and H_1 are F_σ sets in the weak topology.

2 $\implies$ 1 If H_0 and H_1 are disjoint F_σ sets, then we can write $H_0 = \bigcup_{m=1}^{\infty} H_0^m$ and $H_1 = \bigcup_{m=1}^{\infty} H_1^m$ for closed sets H_0^m and H_1^m .

By Theorem 4, there exists for every $m \in \mathbb{N}$, for the pair of hypotheses $(H_0^m, W \setminus H_0^m)$ a strongly consistent FP-test $\varphi_{0,n}^m$, where $\varphi_{0,n}^m = 0$ corresponds to H_0^m and where $\varphi_{0,n}^m = 1$ corresponds to $W \setminus H_0^m$. Similarly, for the pair of hypotheses $(W \setminus H_1^m, H_1^m)$ there exists a strongly consistent FP-test $\varphi_{1,n}^m$ where $\varphi_{1,n}^m = 0$ corresponds to $W \setminus H_1^m$ and $\varphi_{1,n}^m = 1$ corresponds to H_1^m . Define the FP-test φ_n such that

$$\{\varphi_n = 0\} := \bigcup_{m=1}^n \{\varphi_{0,n}^m = 0\} \bigcap_{k=1}^m \{\varphi_{1,n}^k = 0\}$$

$$\{\varphi_n = 1\} := \bigcup_{m=1}^n \{\varphi_{1,n}^m = 1\} \bigcap_{k=1}^m \{\varphi_{0,n}^k = 1\}$$

$$\{\varphi_n = 2\} := \mathcal{X}^n \setminus (\{\varphi_n = 0\} \cup \{\varphi_n = 1\}).$$

For $\mathbb{P} \in H_i$ there is a $m \in \mathbb{N}$ such that $\mathbb{P} \in H_i^m$, and note that $\mathbb{P} \notin H_{1-i}^k$ for all $k \in \mathbb{N}$, so we obtain

$$\begin{aligned} \mathbb{P}^\infty(\liminf_n \{\varphi_n = i\}) &= \mathbb{P}^\infty(\liminf_{n \geq m} \{\varphi_n = i\}) \\ &\geq \mathbb{P}^\infty(\liminf_n \{\varphi_{i,n}^m = i\} \cap \bigcap_{k=1}^m \{\varphi_{1-i,n}^k = i\}) \\ &\geq \sum_{k=1}^m \mathbb{P}^\infty(\liminf_n \{\varphi_{i,n}^m = i\} \cap \{\varphi_{1-i,n}^k = i\}) - m + 1 \geq 1 \end{aligned}$$

by the Fréchet inequality and the consistency of all $\varphi_{i,n}^m$ and $\varphi_{1-i,n}^k$. ■

3.1.3 Proof of Theorem 5

Having Theorem 4 at our disposal, we obtain the following characterisation of testability in terms of both hypotheses being clopen in W .

Proof.

1 $\implies$ 2 By Theorem 4, if H_0 is not closed in W then there is no consistent test with

$$\lim_{n \rightarrow \infty} \sup_{\mathbb{P} \in H_0} \mathbb{P}^n(\varphi_n = 1) = 0$$

and if H_1 is not closed in W then there is no test with

$$\lim_{n \rightarrow \infty} \sup_{\mathbb{P} \in H_1} \mathbb{P}^n(\varphi_n = 0) = 0.$$

2 $\implies$ 1 By Theorem 4 there exists for each $i \in \{0, 1\}$ a strongly consistent FP-test φ_n^i such that $\lim_n \sup_{\mathbb{P} \in H_i} \mathbb{P}^n(\varphi_n^i = 1 - i) = 0$. By defining the FP-test

$$\begin{aligned} \{\varphi_n = 0\} &:= \{\varphi_n^0 = 0\} \cap \{\varphi_n^1 = 0\} \\ \{\varphi_n = 1\} &:= \{\varphi_n^0 = 1\} \cap \{\varphi_n^1 = 1\} \\ \{\varphi_n = 2\} &:= \mathcal{X}^n \setminus (\{\varphi_n = 0\} \cup \{\varphi_n = 1\}), \end{aligned}$$

we have for both $i \in \{0, 1\}$ that $\lim_n \sup_{\mathbb{P} \in H_i} \mathbb{P}^n(\varphi_n = 1 - i) \leq \lim_n \sup_{\mathbb{P} \in H_i} \mathbb{P}^n(\varphi_n^i = 1 - i) = 0$. By Fréchet's inequality we have $\mathbb{P}^\infty(\liminf_n \varphi_n = i) = \mathbb{P}^\infty(\liminf_n \{\varphi_n^0 = i\} \cap \liminf_n \{\varphi_n^1 = i\}) \geq 1$ so φ_n is strongly consistent. ■

3.1.4 Proof of Theorem 6

Proof.

1 $\implies$ 2 Let $d_{BL}(H_0, H_1) > 0$ and assume (7). Let $\gamma < d_{BL}(H_0, H_1)/2$, let $\mathbb{P}_n^x$ denote the empirical measure at $x \in \mathcal{X}^n$, and let

$$\varphi_n(x) := \begin{cases} 0 & \text{if } d_{BL}(\mathbb{P}_n^x, H_0) < \gamma \\ 1 & \text{if } d_{BL}(\mathbb{P}_n^x, H_1) < \gamma \\ 2 & \text{otherwise,} \end{cases}$$

then by weak continuity of $x \mapsto \mathbb{P}_n^x$ the sets $\{\varphi_n = 0\}$ and $\{\varphi_n = 1\}$ are open and disjoint, and we will verify below that φ_n is in fact a BP-test. For $\mathbb{P} \in H_0$ we have $d_{BL}(\mathbb{P}_n^x, H_0) \leq d_{BL}(\mathbb{P}_n^x, \mathbb{P})$, so $\mathbb{P}^n(\varphi_n \neq 0) \leq \mathbb{P}^n(d_{BL}(\mathbb{P}_n^x, \mathbb{P}) \geq \gamma)$. By (7), let $r_n := \sup_{\mathbb{P} \in H_0 \cup H_1} \mathbb{E}_{\mathbb{P}}[d_{BL}(\mathbb{P}_n^X, \mathbb{P})]$, so $r_n \rightarrow 0$. For n large enough that $r_n < \gamma/2$:

$$\begin{aligned} \mathbb{P}^n(\varphi_n \neq 0) &\leq \mathbb{P}^n(d_{BL}(\mathbb{P}_n^x, \mathbb{P}) \geq \gamma) \\ &\leq \mathbb{P}^n(d_{BL}(\mathbb{P}_n^x, \mathbb{P}) - \mathbb{E}_{\mathbb{P}}[d_{BL}(\mathbb{P}_n^X, \mathbb{P})] \geq \gamma/2). \end{aligned}$$

Since for $x = (x_1, \dots, x_n)$ and any i and $x'_i \in \mathcal{X}$ we have with $x' = (x_1, \dots, x'_i, \dots, x_n)$ the bound $|d_{BL}(\mathbb{P}_n^x, \mathbb{P}) - d_{BL}(\mathbb{P}_n^{x'}, \mathbb{P})| \leq 2/n$, McDiarmid's inequality (Van Der Vaart and Wellner, 2023, Proposition 2.15.3) gives

$$\sup_{\mathbb{P} \in H_0} \mathbb{P}^n(\varphi_n \neq 0) \leq \exp(-n\gamma^2/8).$$

The argument for H_1 is identical, giving uniform consistency.

We further show that $\{\varphi_n = 0\}$ and $\{\varphi_n = 1\}$ satisfy the metric-separation condition: there exists $\delta > 0$ such that $d_\infty(\{\varphi_n = 0\}, \{\varphi_n = 1\}) > \delta$ for all $n \in \mathbb{N}$, with d_∞ the ℓ^∞ product metric on $\mathcal{X}^n$. For $x \in \{\varphi_n = 0\}$ and $y \in \{\varphi_n = 1\}$, the triangle inequality gives

$$d_{BL}(\mathbb{P}_n^x, \mathbb{P}_n^y) \geq d_{BL}(H_0, H_1) - d_{BL}(\mathbb{P}_n^x, H_0) - d_{BL}(\mathbb{P}_n^y, H_1) > d_{BL}(H_0, H_1) - 2\gamma > 0.$$

On the other hand, since $|f(x_i) - f(y_i)| \leq d_{\mathcal{X}}(x_i, y_i)$ for $f \in \text{BL}(\mathcal{X}; \mathbb{R})$,

$$d_{BL}(\mathbb{P}_n^x, \mathbb{P}_n^y) = \sup_{f \in \text{BL}} \left| \frac{1}{n} \sum_{i=1}^n (f(x_i) - f(y_i)) \right| \leq \frac{1}{n} \sum_{i=1}^n d_{\mathcal{X}}(x_i, y_i) \leq \max_{1 \leq i \leq n} d_{\mathcal{X}}(x_i, y_i).$$

Combining the two bounds, $d_\infty(x, y) > d_{BL}(H_0, H_1) - 2\gamma$ for any $x \in \{\varphi_n = 0\}$ and $y \in \{\varphi_n = 1\}$, so taking $\delta := d_{BL}(H_0, H_1) - 2\gamma > 0$ gives $d_\infty(\{\varphi_n = 0\}, \{\varphi_n = 1\}) > \delta$ for all $n \in \mathbb{N}$.

2 $\implies$ 3 Let φ_n be a uniformly consistent BP-test, so $\{\varphi_n = 0\}$ and $\{\varphi_n = 1\}$ are metrically separated and thus have disjoint closures. For every $\varepsilon > 0$ there is an $N > 0$ such that $\inf_{\mathbb{P} \in H_i} \mathbb{P}^n(\varphi_n = i) \geq 1 - \varepsilon$ for all $n \geq N$, for $i = 0$ and $i = 1$. Suppose for contradiction that $\overline{H_0} \cap \overline{H_1} \neq \emptyset$, and let $\mathbb{Q}$ be a point in this intersection. Then there are sequences $\mathbb{P}_m^0$ and $\mathbb{P}_m^1$ in H_0 and H_1 respectively converging to $\mathbb{Q}$. Since the map $\mathbb{P} \mapsto \mathbb{P}^n$ is continuous (Billingsley, 1999, Theorem 2.8), we have $(\mathbb{P}_m^i)^n \xrightarrow{w} \mathbb{Q}^n$. Since $\{\varphi_n = 0\}$ and $\{\varphi_n = 1\}$ have disjoint closures, the Portmanteau theorem for closed sets gives

$$\begin{aligned} 1 &\geq \mathbb{Q}^n(\overline{\{\varphi_n = 0\}}) + \mathbb{Q}^n(\overline{\{\varphi_n = 1\}}) \\ &\geq \limsup_m (\mathbb{P}_m^0)^n(\overline{\{\varphi_n = 0\}}) + \limsup_m (\mathbb{P}_m^1)^n(\overline{\{\varphi_n = 1\}}) \\ &\geq \limsup_m (\mathbb{P}_m^0)^n(\{\varphi_n = 0\}) + \limsup_m (\mathbb{P}_m^1)^n(\{\varphi_n = 1\}) \\ &\geq \inf_{\mathbb{P} \in H_0} \mathbb{P}^n(\{\varphi_n = 0\}) + \inf_{\mathbb{P} \in H_1} \mathbb{P}^n(\{\varphi_n = 1\}) \\ &\geq 2(1 - \varepsilon), \end{aligned}$$

so we reach a contradiction for $\varepsilon < 1/2$.

For the final claim, note that if $H_0 \cup H_1$ is precompact, then disjoint closures implies $d_{BL}(H_0, H_1) > 0$ (since the closures are disjoint compact sets). It remains to show that the uniform Glivenko–Cantelli property holds. Since $\mathcal{X}$ is a complete separable metric space, by Prokhorov's theorem W is uniformly tight (Bogachev, 2007, Theorem 8.6.2), so for every $\varepsilon > 0$ there is a compact $K \subseteq \mathcal{X}$ such that $\mathbb{P}(\mathcal{X} \setminus K) \leq \varepsilon$ for all $\mathbb{P} \in W$. This gives

$$d_{BL}(\mathbb{P}_n^x, \mathbb{P}) \leq \sup \left\{ \left| \int_K f d\mathbb{P}_n^x - \int_K f d\mathbb{P} \right| : f \in \text{BL} \right\} + \mathbb{P}_n^x(\mathcal{X} \setminus K) + \mathbb{P}(\mathcal{X} \setminus K),$$

so $\mathbb{E}[d_{BL}(\mathbb{P}_n^x, \mathbb{P})] \leq \mathbb{E}[\sup \{ |\int_K f d\mathbb{P}_n^x - \int_K f d\mathbb{P}| : f \in \text{BL} \}] + 2\varepsilon$. Since $\text{BL}(\mathcal{X}; \mathbb{R})$ is equicontinuous and bounded, by Ascoli's theorem (Munkres, 2014, Theorem 47.1) it is precompact in the topology of uniform convergence on compacta. Restricted to K , this gives precompactness of $\text{BL}(K; \mathbb{R})$ in the sup-norm, hence for every $\varepsilon > 0$ there is a N_ε and functions $f_1, \dots, f_{N_\varepsilon} \in \text{BL}(K; \mathbb{R})$ such that for every $f \in \text{BL}(K; \mathbb{R})$, $\|f - f_i\| \leq \varepsilon$ for some i . The triangle inequality then gives

$$\sup \left\{ \left| \int_K f d\mathbb{P}_n^x - \int_K f d\mathbb{P} \right| : f \in \text{BL} \right\} \leq \max \left\{ \left| \int_K f_i d\mathbb{P}_n^x - \int_K f_i d\mathbb{P} \right| : i = 1, \dots, N_\varepsilon \right\} + 2\varepsilon.$$

By Jensen's inequality applied to the square we obtain

$$\begin{aligned} \mathbb{E} \left[\left| \int_K f_i d\mathbb{P}_n^x - \int_K f_i d\mathbb{P} \right|^2 \right] &= \mathbb{E} \left[\left| \frac{1}{n} \sum_{j=1}^n \mathbb{1}_K(X_j) f_i(X_j) - \mathbb{E}_{\mathbb{P}}[\mathbb{1}_K f_i] \right|^2 \right] \\ &\leq \sqrt{\text{Var} \left(\frac{1}{n} \sum_{j=1}^n \mathbb{1}_K(X_j) f_i(X_j) \right)} \leq \frac{1}{\sqrt{n}}, \end{aligned}$$

so by combining these intermediate results we obtain

$$\mathbb{E}_{\mathbb{P}} [d_{BL}(\mathbb{P}_n^X, \mathbb{P})] \leq \frac{N_{\varepsilon}}{\sqrt{n}} + 4\varepsilon,$$

so (7) holds. ■

4 The hardness of conditional independence testing

We now turn to applications of the preceding results to conditional independence testing. Lauritzen (2024) shows that conditional independence is closed under limits in the total variation metric. However, in general, weak convergence does not preserve conditional independence: for a weakly convergent sequence $\mathbb{P}_n(X, Y, Z) \xrightarrow{w} \mathbb{P}(X, Y, Z)$ with $X \perp\!\!\!\perp_{\mathbb{P}_n} Y \mid Z$ for all $n \in \mathbb{N}$, we might have $X \not\perp\!\!\!\perp_{\mathbb{P}} Y \mid Z$.

Example 2 (Lauritzen, 1996, Example 3.11). Consider a trivariate Gaussian $\mathbb{P}_n(X, Y, Z)$ with mean zero, and (conditional) covariance matrices given by

$$\Sigma^n = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{\sqrt{n}} \\ \frac{1}{2} & 1 & \frac{1}{\sqrt{n}} \\ \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \frac{2}{n} \end{pmatrix} \quad \text{and} \quad \Sigma_{XY|Z}^n = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix},$$

so we have conditional independence $X \perp\!\!\!\perp_{\mathbb{P}_n} Y \mid Z$. Since $\mathbb{P}_n(X, Y, Z) \xrightarrow{w} \mathbb{P}(X, Y, Z)$ with $\mathbb{P}(X, Y, Z)$ a degenerate multivariate Gaussian distribution with (conditional) covariance matrices

$$\Sigma = \begin{pmatrix} 1 & \frac{1}{2} & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \Sigma_{XY|Z} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix},$$

we have $X \not\perp\!\!\!\perp_{\mathbb{P}} Y \mid Z$, so conditional independence is not maintained under weak limits. Other examples can be found in Barbie and Gupta (2014) and Saldi and Yüksel (2022).

Combining this example with Theorem 4, we immediately obtain that for real-valued random variables, conditional independence is not consistently FP-testable with uniform error control under the null. We strengthen this result by allowing for rather general sample spaces, and showing that consistent FP-tests don't exist for conditional independence. We use the following lemmas:

Lemma 4. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be complete separable metric spaces with $\mathcal{Z}$ perfect⁷ then $H_0 := \{\mathbb{P} : X \perp\!\!\!\perp_{\mathbb{P}} Y \mid Z\}$ is dense in $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$.*

Proof. Probability measures of the form $\mathbb{P} = \sum_{i=1}^n a_i \delta_{k_i}$ with $k_i = (x_i, y_i, z_i) \in \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$ are dense in $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ (Bogachev, 2007, Example 8.1.6). If there are $i \neq j$ such that $z_i = z_j$, then for every $\varepsilon > 0$ there is a $z'_j \in \mathcal{Z}$ such that $z'_j \neq z_i$ for all $i = 1, \dots, n$ and $d(z_j, z'_j) < \varepsilon/a_j$. Defining $k'_j := (x_j, y_j, z'_j)$ and $\mathbb{P}' := \sum_{i=1}^n a_i \delta_{k_i} + a_j(\delta_{k'_j} - \delta_{k_j})$ we have that $d_{BL}(\mathbb{P}, \mathbb{P}') < \varepsilon$. Hence, convex combinations of point-masses with distinct Z -coordinates are dense in $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$. For any such $\mathbb{P}'$ and any z_i in the support of $\mathbb{P}'(Z)$ we have $\mathbb{P}'(X, Y \mid Z = z_i) = \delta_{(x_i, y_i)} = \delta_{x_i} \delta_{y_i}$, so $X \perp\!\!\!\perp_{\mathbb{P}'} Y \mid Z$. ■

⁷That is, $\mathcal{Z}$ has no isolated points. Since $\mathcal{Z}$ is also Polish, it is uncountable (Kechris, 1995, Corollary 6.3).

Lemma 5. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be complete separable metric spaces, then the set $H_1 := \{\mathbb{P} : X \not\perp_{\mathbb{P}} Y \mid Z\}$ is dense in $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$.*

Proof. In Boeken et al. (2026), Corollary 1 it is shown that $H_1 := \{\mathbb{P} : X \not\perp_{\mathbb{P}} Y \mid Z\}$ is dense in $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ with respect to the total variation metric. Since $d_{BL} \leq d_{TV}$, the assertion holds. ■

Note that in general the two hypotheses can be dense and consistently testable, see e.g. Example 1 where we provide a consistent test for the pair of hypotheses $H_0 = [0, 1] \cap \mathbb{Q}$ and $H_1 = [0, 1] \cap (\mathbb{Q} + \sqrt{2})$. However, this only occurs in special cases, since H_0 and H_1 being dense and F_σ implies that W is meager in itself. By the Baire Category Theorem the space $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ is not meager in itself, so we obtain the following impossibility result.

Theorem 7. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be complete separable metric spaces with $\mathcal{Z}$ perfect, then the hypotheses $H_0 := \{\mathbb{P} : X \perp_{\mathbb{P}} Y \mid Z\}$ and $H_1 := \{\mathbb{P} : X \not\perp_{\mathbb{P}} Y \mid Z\}$ are not consistently FP-testable.*

Proof. Since $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ equipped with the weak topology is complete, the Baire Category Theorem implies that it is not meager in itself. By Lemmas 4 and 5 we have that H_0 and H_1 are dense. If H_0 and H_1 are both F_σ , then they are both G_δ (i.e. a countable intersection of open sets). The complement of a dense G_δ set is meager, implying that H_0 and H_1 are meager. The space $W = H_0 \cup H_1$ would then be meager as well giving a contradiction, so H_0 and H_1 cannot both be F_σ . From Theorem 3 we conclude that H_0, H_1 are not consistently FP-testable. ■

The preceding result is for example applicable when X, Y, Z take values in (complete separable metric) function spaces such as $C([0, 1], \mathbb{R}^d)$ (Manten et al., 2024), $L^p([0, 1], \mathbb{R}^d)$ (Lundborg et al., 2022) or the Skorohod space $\mathbb{D}([0, 1], \mathbb{R}^d)$ (Boeken and Mooij, 2024), for example when they represent measurements of continuous-time stochastic processes.

5 Weak closedness of conditional independence

Having established that conditional independence is generally not FP-testable, we will now search for conditions under which conditional independence is closed, F_σ , or metrically separated, implying the existence of a (uniformly) consistent FP-test or BP-test. In particular, we will show that if $\mathbb{P}_n(X, Y, Z) \xrightarrow{w} \mathbb{P}(X, Y, Z)$ and the conditional distributions $z \mapsto \mathbb{P}_n(X \mid Z = z)$ share a modulus of continuity, then conditional independence $X \perp Y \mid Z$ is maintained in the limit. First, we formalise what is meant by a conditional distribution having a modulus of continuity.

A Markov kernel $\mathbb{P}(X \mid Z)$ is a measurable map $\mathcal{Z} \rightarrow \mathcal{P}(\mathcal{X})$.⁸ For Markov kernels $\mathbb{P}(X \mid Y), \mathbb{P}(Y \mid Z)$, their *product* is defined as the Markov kernel

$$\mathbb{P}(X \mid Y) \otimes \mathbb{P}(Y \mid Z) : \mathcal{Z} \rightarrow \mathcal{P}(\mathcal{X} \times \mathcal{Y}), \quad z \mapsto \left(D \mapsto \int_D d\mathbb{P}(x \mid y) d\mathbb{P}(y \mid z) \right)$$

where $D \in \mathcal{B}(\mathcal{X} \times \mathcal{Y})$. If $\mathcal{X}$ is a complete separable metric space, then for a given joint distribution $\mathbb{P}(X, Z)$ there exists a Markov kernel (a version of the *conditional distribution*) $\mathbb{P}(X \mid Z)$ such that $\mathbb{P}(X, Z) = \mathbb{P}(X \mid Z) \otimes \mathbb{P}(Z)$. This Markov kernel is $\mathbb{P}(Z)$ -almost everywhere uniquely defined.

Throughout, fix a non-decreasing function $\omega : [0, \infty) \rightarrow [0, \infty)$ with $\omega(0) = 0$, continuous at 0, and satisfying $\liminf_{t \downarrow 0} \omega(t)/t > 0$.⁹ Our main focus is on distributions $\mathbb{P}(X, Y, Z)$ for which there exists a version of the conditional distribution $\mathbb{P}(X \mid Z)$ that has *modulus of continuity* ω , i.e.

$$d_{BL}(\mathbb{P}(X \mid Z = z), \mathbb{P}(X \mid Z = z')) \leq \omega(d_{\mathcal{Z}}(z, z'))$$

⁸Here, $\mathcal{P}(\mathcal{X})$ is equipped with the Borel σ -algebra generated by the weak topology, which coincides with the smallest σ -algebra that makes for all $D \in \mathcal{B}(\mathcal{X})$ the evaluation map $\mathbb{P} \mapsto \mathbb{P}(X \in D)$ measurable (Ghosal and Van Der Vaart, 2017, Proposition A.5). This definition of a Markov kernel is equivalent to the common definition that for all $D \in \mathcal{B}(\mathcal{X})$ the map $z \mapsto \mathbb{P}(X \in D \mid Z = z)$ is measurable, and for every $z \in \mathcal{Z}$ the map $D \mapsto \mathbb{P}(X \in D \mid Z = z)$ is a probability measure.

⁹Moduli with $\lim_{t \downarrow 0} \omega(t)/t = 0$ would force any ω -equicontinuous function on a path-connected metric space to be constant, and are excluded.

for all $z, z' \in \mathcal{Z}$, where $d_{\mathcal{Z}}$ denotes the metric on $\mathcal{Z}$ – we will always take this continuous version of the conditional distribution, which is uniquely determined on the support of $\mathbb{P}(Z)$. Special cases are L -Lipschitz Markov kernels for $\omega(t) = Lt$, Hölder-continuous Markov kernels for $\omega(t) = Lt^\alpha$ and $\alpha \in (0, 1]$, log-Hölder Markov kernels for $\omega(t) = L/\log(1/t)^\alpha$, and other regularity classes. It follows from the definitions that the Markov kernel $z \mapsto \mathbb{P}(X | Z = z)$ has modulus of continuity ω if and only if the conditional expectation $z \mapsto \mathbb{E}[f(X) | Z = z]$ has modulus of continuity ω for all $f \in \text{BL}(\mathcal{X}; \mathbb{R})$.

First, we have that the modulus-of-continuity assumption is closed in the weak topology:

Theorem 8. *Let $\mathcal{X}, \mathcal{Z}$ be complete separable metric spaces with $\mathcal{Z}$ locally compact. The set $W_\omega := \{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Z}) : \mathbb{P}(X | Z) \text{ has modulus of continuity } \omega\}$ is closed in the weak topology.*

Further, a shared modulus of continuity for the conditional $\mathbb{P}(X | Z)$ or $\mathbb{P}(Y | Z)$ implies that conditional independence is closed in the weak topology.

Theorem 9. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be complete separable metric spaces with $\mathcal{Z}$ locally compact. If $\mathbb{P}_n(X, Y, Z) \xrightarrow{w} \mathbb{P}(X, Y, Z)$ where $X \perp_{\mathbb{P}_n} Y | Z$ and $\mathbb{P}_n(X | Z)$ has modulus of continuity ω for all $n \in \mathbb{N}$, then $X \perp_{\mathbb{P}} Y | Z$.*

The proofs of Theorems 8 and 9 are given in Appendix A. The local compactness hypothesis on $\mathcal{Z}$ is satisfied for instance by Euclidean spaces $\mathbb{R}^d$ and discrete spaces; it excludes infinite-dimensional Banach spaces such as ℓ^p or $C([0, 1], \mathbb{R})$. No local compactness is required of $\mathcal{X}$ (or $\mathcal{Y}$ in Theorem 9), so X (c.q. Y) may take values in such function spaces.

5.1 Sufficient conditions for Markov kernels with modulus of continuity

In Section 6 we will use this modulus-of-continuity assumption for conditional independence testing. To make it better accessible, we investigate sufficient conditions for this assumption to hold.

Scheffé's theorem implies that if a conditional density is continuous in its conditioning variable, so $p(x | z_n) \rightarrow p(x | z)$ for some sequence $z_n \rightarrow z$, then $\mathbb{P}(X | Z = z_n) \xrightarrow{w} \mathbb{P}(X | Z = z)$. The following proposition contains a uniform-continuity analogue of this result.

Proposition 1. *Markov kernels with modulus of continuity have the following properties:*

- a) *If $\mathbb{P}(X | Z)$ has a density $p(x | z)$ with respect to a finite measure $\mathbb{Q}(X \in \mathcal{X}) \leq M < \infty$ such that $z \mapsto p(x | z)$ has modulus of continuity ω uniformly in x , then $\mathbb{P}(X | Z)$ has modulus of continuity $M\omega$.*
- b) *If $\mathcal{Z}$ is discrete, then $\mathbb{P}(X | Z)$ is 2-Lipschitz.*
- c) *If $z \mapsto \mathbb{P}(X \in D | Z = z)$ has modulus of continuity ω for all $D \in \mathcal{B}(\mathcal{X})$, then $z \mapsto \mathbb{P}(X | Z = z)$ has modulus of continuity ω as well.*
- d) *If $\mathbb{P}(X, Z) \sim \mathcal{N}(\mu, \Sigma)$ is multivariate Gaussian, then $\mathbb{P}(X | Z)$ has Lipschitz constant $\|\Sigma_{XZ}\Sigma_{ZZ}^{-1}\|_{\text{op}}$.¹⁰*

Proof.

- a) For any $f \in \text{BL}(\mathcal{X}; \mathbb{R})$ and $z, z' \in \mathcal{Z}$ we have $|\mathbb{E}[f(X) | z] - \mathbb{E}[f(X) | z']| \leq \int_{\mathcal{X}} |p(x | z) - p(x | z')| d\mathbb{Q}(x) \leq \omega(d_{\mathcal{Z}}(z, z')) \mathbb{Q}(\mathcal{X}) \leq M\omega(d_{\mathcal{Z}}(z, z'))$.
- b) For any $f \in \text{BL}(\mathcal{X}; \mathbb{R})$ and $z, z' \in \mathcal{Z}$ with $z \neq z'$ we have $|\int f(x) d(\mathbb{P}(x | z) - \mathbb{P}(x | z'))| \leq 2 = 2d_{\mathcal{Z}}(z, z')$, since the discrete metric satisfies $d_{\mathcal{Z}}(z, z') = 1$ whenever $z \neq z'$.
- c) This follows directly from the definition of the total variation metric, which upper-bounds the bounded Lipschitz metric.

¹⁰Here, $\|\cdot\|_{\text{op}}$ denotes the operator norm.

- d) Given $\mathbb{P}(X, Z) \sim \mathcal{N}(\mu, \Sigma)$, the conditional $\mathbb{P}(X | Z = z)$ is Gaussian with mean $\mu_{X|Z}(z) := \mu_X + \Sigma_{XZ}\Sigma_{ZZ}^{-1}(z - \mu_Z)$ and covariance matrix $\Sigma_{X|Z} := \Sigma_{XX} - \Sigma_{XZ}\Sigma_{ZZ}^{-1}\Sigma_{ZX}$. Let $U \sim \mathcal{N}(0, \Sigma_{X|Z})$. For any $f \in \text{BL}(\mathcal{X}; \mathbb{R})$ we have

$$\begin{aligned} |\mathbb{E}[f(X) | Z = z] - \mathbb{E}[f(X) | Z = z']| &= |\mathbb{E}[f(\mu_{X|Z}(z) + U)] - \mathbb{E}[f(\mu_{X|Z}(z') + U)]| \\ &\leq \mathbb{E}[|f(\mu_{X|Z}(z) + U) - f(\mu_{X|Z}(z') + U)|] \\ &\leq \|\mu_{X|Z}(z) - \mu_{X|Z}(z')\| \\ &\leq \|\Sigma_{XZ}\Sigma_{ZZ}^{-1}\|_{\text{op}}\|z - z'\|. \end{aligned}$$

One readily verifies that the conditional distributions $\mathbb{P}_n(X | Z)$ and $\mathbb{P}_n(Y | Z)$ from Example 2 are $\sqrt{n}/2$ -Lipschitz and that they satisfy $d_{BL}(\mathbb{P}_n(X | z), \mathbb{P}_n(X | z + 1/\sqrt{n})) > 1/3$ for every $z \in \mathbb{R}$ and $n \in \mathbb{N}$ (by taking the test function $f(x) = \sin(x)$), so the family $\{\mathbb{P}_n(X | Z)\}_n$ admits no shared modulus of continuity. Hence, the fact that conditional independence is not maintained in the limit hints at the sharpness of the modulus condition in Theorem 9. ■

5.2 Weak closedness of conditional independence via total variation

We can also take a different approach to finding sufficient conditions under which conditional independence is weakly closed, which does not use uniform continuity of the Markov kernels. Namely, by Lauritzen (2024), conditional independence is closed in total variation. For sets of probability measures for which the total variation topology and weak topology coincide, we then immediately obtain that conditional independence is weakly closed. To this end, we consider classes of measures with well-behaved densities. Let μ be a locally finite Borel measure on complete separable metric space $\mathcal{X}$, let ω be a modulus of continuity, let $M > 0$, and let $W_{\mu, \omega, M}$ be the class of distributions with a density p with respect to μ that are ω -equicontinuous and uniformly bounded by M , that is, we have $|p(x) - p(y)| < \omega(|x - y|)$ and $p(x) \leq M$ for all $x, y \in \mathcal{X}$. Extending Boos (1985) to complete separable metric spaces, Boeken et al. (2026) (Lemma 6) show that the total variation topology and weak topology coincide on $W_{\mu, \omega, M}$, and that $W_{\mu, \omega, M}$ is closed in the weak topology. By Lauritzen (2024), this implies the following result:

Theorem 10. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be complete separable metric spaces, and let μ, M, ω be given. If $\mathbb{P}_n(X, Y, Z) \xrightarrow{w} \mathbb{P}(X, Y, Z)$ with $X \perp_{\mathbb{P}_n} Y | Z$ and $\mathbb{P}_n \in W_{\mu, \omega, M}$ for all $n \in \mathbb{N}$, then $\mathbb{P} \in W_{\mu, \omega, M}$ as well and $X \perp_{\mathbb{P}} Y | Z$.*

Another approach would be to consider sufficiently regular exponential families, for which by Barndorff-Nielsen (2014), Section 8.1, Theorem 8.3, the parameter space with the Euclidean topology is homeomorphic to the set of probability measures with the weak topology. By Scheffé's theorem, this topology then coincides with the total variation topology.

The hypotheses of Theorems 9 and 10 are complementary, neither implying the other. Also, Theorem 9 cannot be proved by invoking Lauritzen (2024): weak convergence plus the modulus-of-continuity assumption does not imply total variation convergence (for example, let $\mathbb{P}_n(X) = \delta_{1/n}$ and $\mathbb{P}_n(Y | X = x) = \delta_x$, which is 1-Lipschitz and $\mathbb{P}_n(Y | X) \otimes \mathbb{P}_n(X)$ converges weakly to the point mass on $(0, 0)$, but not in total variation).

5.3 Related literature

As sufficient condition for conditional independence to be closed in the weak topology, Jordan (1977) and Hellwig (1996) considered probability measures $\mathbb{P}$ with $z \mapsto \mathbb{P}(X | Z = z)$ continuous, but Barbie and Gupta (2014) gave a counterexample. They show that conditional independence is closed in the *topology of information* (see also Backhoff-Veraguas et al. (2020)), and show that this topology coincides with the weak topology under similar equicontinuity conditions as Theorem 9.

6 Sufficient conditions for conditional independence testing

Having the topological sufficient conditions for testability from Section 3 and the sufficient conditions for conditional independence to be weakly closed from Section 5 at our disposal, we can relatively easily prove the existence of consistent conditional independence FP-tests, in various settings.

To this end, recall that $W_\omega := \{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}) : \mathbb{P}(X | Z) \text{ has modulus of continuity } \omega\}$. Also, recall the definition of $W_{\mu,\omega,M}$ from Section 5.2.

Theorem 11. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be complete separable metric spaces, let $W \subseteq \mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ and consider the hypotheses $H_0 := \{\mathbb{P} \in W : X \perp_{\mathbb{P}} Y | Z\}$ and $H_1 := \{\mathbb{P} \in W : X \not\perp_{\mathbb{P}} Y | Z\}$.*

- a) *If $\mathcal{Z}$ is locally compact and $W \subseteq W_\omega$ for some modulus of continuity ω , then there exists a strongly consistent FP-test with uniform error control under H_0 .*
- b) *If $\mathcal{Z}$ is discrete and $W \subseteq \mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$, then there exists a strongly consistent FP-test with uniform error control under H_0 .*
- c) *If $W \subseteq W_{\mu,\omega,M}$ for some (μ, ω, M) , then there exists a strongly consistent FP-test with uniform error control under H_0 .*
- d) *If $\mathcal{Z}$ is locally compact and $W \subseteq W_\omega$ is precompact or satisfies the uniform Glivenko–Cantelli property, and we consider for given $\varepsilon > 0$ the alternative hypothesis*

$$H_1^\varepsilon := \{\mathbb{P} \in W : d_{BL}(\mathbb{P}(X | Z) \otimes \mathbb{P}(Y, Z), \mathbb{P}(X, Y, Z)) \geq \varepsilon\},$$

then there exists a uniformly consistent BP-test.

- e) *If $\mathcal{Z}$ is locally compact and $W \subseteq W_\infty := \bigcup_{n \in \mathbb{N}} W_{\omega_n}$ for some countable family of moduli $\{\omega_n\}_{n \in \mathbb{N}}$, then there exists a strongly consistent FP-test.*
- f) *If $W \subseteq W_\infty^{TV} := \bigcup_{n \in \mathbb{N}} W_{\mu_n, \omega_n, n}$ for some countable family $\{(\mu_n, \omega_n)\}_{n \in \mathbb{N}}$, then there exists a strongly consistent FP-test.*

Proof.

- a) By Theorem 9 we have that H_0 is closed in $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$, hence also in W_ω . The result follows from Theorem 4.
- b) If $\mathcal{Z}$ is discrete, then by Proposition 1.b) we have that $W_\omega = \mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ for $\omega(t) = 2t$, and $\mathcal{Z}$ is locally compact (every singleton is open and compact), so the result follows from part a).
- c) By Theorem 10, H_0 is closed in $W_{\mu,\omega,M}$, so the result follows from Theorem 4.
- d) From Theorem 9 it follows that the map $T : \mathbb{P}(X, Y, Z) \mapsto \mathbb{P}(X | Z) \otimes \mathbb{P}(Y, Z)$ from W_ω to $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ is continuous. Hence, the map

$$f : W_\omega \mapsto [0, 1], \quad \mathbb{P}(X, Y, Z) \mapsto d_{BL}(\mathbb{P}(X | Z) \otimes \mathbb{P}(Y, Z), \mathbb{P}(X, Y, Z))$$

is continuous as well, so $H_0 \subseteq f^{-1}(\{0\})$ and $H_1^\varepsilon \subseteq f^{-1}([\varepsilon, 1])$ are contained in disjoint closed subsets of W_ω . Since W_ω is closed in $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ by Theorem 8, H_0 and H_1^ε are closed in $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ as well. If W is precompact or satisfies the uniform Glivenko–Cantelli property, the result follows from Theorem 6.

- e) For each $n \in \mathbb{N}$, the set W_{ω_n} is closed in $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ by Theorem 8, hence also in W_∞ . Writing $H'_0 = \{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}) : X \perp_{\mathbb{P}} Y | Z\}$, we have that $H'_0 \cap W_{\omega_n}$ is closed in W_{ω_n} by Theorem 9, hence also in W_∞ . This gives that $\bigcup_{n \in \mathbb{N}} H'_0 \cap W_{\omega_n}$ is F_σ in W_∞ and hence $H_0 \subseteq \bigcup_{n \in \mathbb{N}} H'_0 \cap W_{\omega_n}$ is F_σ in W . Similarly, writing $H'_1 = \{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}) : X \not\perp_{\mathbb{P}} Y | Z\}$ we have that $H'_1 \cap W_{\omega_n}$ is open and hence F_σ in W_{ω_n} , so by a similar reasoning H_1 is F_σ in W , and the result follows from Theorem 3.

f) By Theorem 10, each set $W_{\mu_n, \omega_n, n}$ is closed in $\mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ and $H'_0 \cap W_{\mu_n, \omega_n, n}$ is closed in $W_{\mu_n, \omega_n, n}$, so the result follows from a similar proof as part e). ■

Note that by symmetry, one can also let $\mathbb{P}(Y | Z)$ have modulus of continuity ω in the definition of W_ω as considered in Theorem 11.

Remark 1. Note that the hypotheses $H_0 := W_\omega$ and $H_1 := \mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}) \setminus W_\omega$ are also FP-testable, since W_ω is closed by Theorem 8. Hence, these regularity assumptions for conditional independence testing *are themselves testable*. An assumption-free consistent FP-test with uniform error control under H_0 therefore exists for the hypotheses $H_0 := \{\mathbb{P} \in W_\omega : X \perp\!\!\!\perp Y | Z\}$ and $H_1 := \mathcal{P}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}) \setminus H_0$. The same reasoning applies to $W_{\mu, \omega, M}$.

6.1 Related literature

Warren (2021) proposes a conditional independence test based on binning the space $\mathcal{Z}$, and proves that it is pointwise asymptotically valid under the null and consistent under the alternative, under the assumption that the distributions $\mathbb{P}(X, Y | Z)$, $\mathbb{P}(X | Z)$ and $\mathbb{P}(Y | Z)$ are L -Lipschitz maps from a compact space $\mathcal{Z}$ to the space of probability measures equipped with the p -Wasserstein distance W_p . To compare this assumption to the equicontinuity assumption that we make, we have $d_{BL}(\mathbb{P}_0, \mathbb{P}_1) \leq W_p(\mathbb{P}_0, \mathbb{P}_1)$, with equality $d_{BL}(\mathbb{P}_0, \mathbb{P}_1) = W_1(\mathbb{P}_0, \mathbb{P}_1)$ if $\mathbb{P}_0$ and $\mathbb{P}_1$ have bounded support (Bogachev, 2007, Theorem 8.10.45). Theorem 11.a) shows that, under the assumptions of Warren (2021), there even exists a test that is valid at every sample size $n \in \mathbb{N}$, and one requires only $\mathbb{P}(X | Z)$ or $\mathbb{P}(Y | Z)$ to admit a modulus of continuity ω .

Neykov et al. (2021) propose a conditional independence test which obtains a minimax-optimal rate, under the assumption that X, Y, Z are compactly supported, have densities, and $\mathbb{P}(X | Z)$ and $\mathbb{P}(Y | Z)$ are L -Lipschitz maps from $\mathcal{Z}$ to the space of probability measures equipped with the total variation metric.

Györfi and Walk (2012) propose a conditional independence test and aim at proving its strong consistency without any regularity conditions on H_0 and H_1 , but Neykov et al. (2021) point out a mistake in their proof. Dai and Song (2025) propose a consistent test, under the assumption of a smooth parametric model. For data beyond real-valued random variables, there is some work about conditional independence testing where X, Y, Z take values in function spaces, representing measurements of continuous-time stochastic processes. Lundborg et al. (2022) propose an asymptotically valid test under the assumption that $\mathbb{E}[X | Z]$ and $\mathbb{E}[Y | Z]$ can be estimated sufficiently well, and Manten et al. (2024) propose a weakly consistent test under the assumption that $\mathbb{E}[X | Z]$, $\mathbb{E}[Y | Z]$ and $\mathbb{E}[(X, Y) | Z]$ can be estimated with certain kernel methods. None of these tests take any regularity of the critical region into account.

7 Discussion

This work establishes a topological framework for understanding the testability of statistical hypotheses under the constraint of finite-precision measurements. Our results reveal that the feasibility of constructing consistent tests with various types of error control can be characterized entirely in terms of the topological properties (F_σ , closed, clopen, or metric separation) of the null and alternative hypotheses in the weak topology on the space of probability measures. An important implication is the non-testability of conditional independence hypotheses in general: because both the null and alternative hypotheses are dense and their union is a Baire space, no consistent FP-test exists without additional assumptions. This generalizes and strengthens prior results in the literature, and underscores that conditional independence — while foundational in causal inference and graphical models — is not a testable property without regularity conditions. By imposing a shared modulus-of-continuity assumption on the conditional distributions, we recover testability: we show that this assumption ensures closedness in the weak topology, enabling the construction of tests with uniform error control under H_0 . This yields sufficient conditions — including Lipschitz, Hölder, and log-Hölder regularity classes as special cases — under which conditional independence becomes statistically testable. In particular, we prove that the regularity assumption itself is testable.

Despite these contributions, several important questions remain open. While the established topological conditions are necessary and sufficient, they might not directly provide one with an explicit test: given an open H_1 , one must find open $A_{ij} \subseteq \mathcal{X}$ and $q_{ij} \in [0, 1]$ such that $H_1 = \bigcup_{i=1}^{\infty} \bigcap_{j=1}^{m_i} \{\mathbb{P} : \mathbb{P}(A_{ij}) > q_{ij}\}$ in order to construct a test, and this representation may be difficult to identify in practice. It would be interesting to find such an explicit representation for conditional independence. Beyond the existence questions addressed here, the rates of error decay attainable by FP- and BP-tests, and whether our constructions are (minimax) optimal in any sense, remain open. In the setting of Theorem 6, it would be interesting to study in more practical applications the tradeoff between the level of bounded precision, the separation of the hypotheses, and the rate of uniform convergence. Another interesting avenue for future research would be to allow for dependent data sampling. By Theorem 2, uniform error control under H_0 can be interpreted as $\sup_{\mathbb{P} \in H_0} \mathbb{P}^\infty(\exists n : \varphi_n \neq 0) \leq \alpha$, which is closely related to anytime-valid p-values, e-values and e-processes (Wang and Ramdas, 2022; Grünwald et al., 2024). Since exchangeable sequences of data can be obtained as convex combinations of i.i.d. processes (Ramdas et al., 2022), we conjecture that our results can be extended to anytime-valid tests for exchangeable data. Extensions to topological characterisations of the existence of e-values and e-processes for arbitrary sampling schemes would be of significant interest. It is plausible that analogous topological criteria govern the possibility of valid online or interactive hypothesis testing in complex models. It remains an open question whether there exists a consistent FP-test for conditional independence if one merely assumes that the distribution has a density. Finally, from Example 2 it seems that some kind of equicontinuity of $\mathbb{P}(X | Z)$ or $\mathbb{P}(Y | Z)$ is necessary for conditional independence testing with uniform error control under H_0 . Finding these necessary conditions would be of great interest.

In summary, our results offer a unified topological perspective on the fundamental limits of statistical inference under measurement constraints. They resolve open problems around the testability of nonparametric hypotheses, while opening avenues for further exploration in both theoretical and applied statistics.

A Proofs of Theorems 8 and 9

In this appendix, we prove Theorems 8 and 9 from Section 5.

Lemma 6. *Let $\mathcal{X}, \mathcal{Z}$ be complete separable metric spaces with $\mathcal{Z}$ locally compact. If $\mathbb{P}_n(X, Z) \xrightarrow{w} \mathbb{P}(X, Z)$ and $\mathbb{P}_n(X | Z)$ has modulus of continuity ω for all $n \in \mathbb{N}$, then $\mathbb{P}(X | Z)$ admits a version with modulus of continuity ω on $\mathcal{Z}$, and $\mathbb{P}_n(X | Z) \rightarrow \mathbb{P}(X | Z)$ uniformly on compact subsets of $\mathcal{Z}$.*

Proof. The shared modulus of continuity implies that $\{z \mapsto \mathbb{E}_n[f(X) | Z = z]\}_{n \in \mathbb{N}}$ is equicontinuous on $\mathcal{Z}$ for every $f \in \text{BL}_1(\mathcal{X}; \mathbb{R})$. By Sweeting (1989), Theorem 4, applied with $V = \mathcal{Z}$, this equicontinuity together with the joint convergence $\mathbb{P}_n(X, Z) \xrightarrow{w} \mathbb{P}(X, Z)$ gives $\mathbb{P}_n(X | Z = z) \xrightarrow{w} \mathbb{P}(X | Z = z)$ for every $z \in \mathcal{Z}$, where $\mathbb{P}(X | Z = z)$ is the unique continuous version of the conditional distribution.¹¹

For each $z \in \mathcal{Z}$, pointwise weak convergence and Prokhorov's theorem (Bogachev, 2007, Theorem 8.6.2) imply that $\{\mathbb{P}_n(X | Z = z) : n \in \mathbb{N}\}$ is relatively compact in $(\mathcal{P}(\mathcal{X}), d_{BL})$. Combined with the equicontinuity, the family $\{z \mapsto \mathbb{P}_n(X | z)\}_{n \in \mathbb{N}} \subseteq C(\mathcal{Z}, \mathcal{P}(\mathcal{X}))$ is equicontinuous and pointwise relatively compact, so by Ascoli's theorem (Munkres, 2014, Theorem 47.1) it has compact closure in $C(\mathcal{Z}, \mathcal{P}(\mathcal{X}))$ equipped with the topology of uniform convergence on compacta. Since $\mathcal{Z}$ is locally compact and second countable, the topology of uniform convergence on compact subsets of $\mathcal{Z}$ is metrizable (Munkres, 2014, Exercise 10 in §46). Relative compactness in a metrizable space is equivalent to sequential relative compactness, so every subsequence (n_k) has a further subsequence (n_{k_j}) converging uniformly on compacta of $\mathcal{Z}$ to some continuous limit $g : \mathcal{Z} \rightarrow \mathcal{P}(\mathcal{X})$. By the pointwise convergence $\mathbb{P}_n(X | Z = z) \xrightarrow{w} \mathbb{P}(X | Z = z)$ at every $z \in \mathcal{Z}$, this limit must satisfy $g(z) = \mathbb{P}(X | Z = z)$ for all $z \in \mathcal{Z}$. Since every subsequence has a further subsequence converging to the same limit $\mathbb{P}(X | Z = \cdot)$, the whole sequence converges to it: $\sup_{z \in K} d_{BL}(\mathbb{P}_n(X | z), \mathbb{P}(X | z)) \rightarrow 0$ for every compact $K \subseteq \mathcal{Z}$. The

¹¹Although Sweeting (1989), Theorem 4 is stated under the hypothesis that $\mathcal{X}$ is locally compact, the direction (i) $\Rightarrow$ (ii) used here is established via Sweeting (1989), Theorem 3 on the locally compact space $\mathcal{Z}$ and does not require local compactness of $\mathcal{X}$.

limit map $z \mapsto \mathbb{P}(X | Z = z)$ inherits modulus of continuity ω on $\mathcal{Z}$ as a uniform-on-compacta limit of ω -equicontinuous maps. $\blacksquare$

Theorem 8. *Let $\mathcal{X}, \mathcal{Z}$ be complete separable metric spaces with $\mathcal{Z}$ locally compact. The set $W_\omega := \{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Z}) : \mathbb{P}(X | Z) \text{ has modulus of continuity } \omega\}$ is closed in the weak topology.*

Proof. Let $\mathbb{P}_n(X, Z) \xrightarrow{w} \mathbb{P}(X, Z)$ with $\mathbb{P}_n(X | Z)$ having modulus of continuity ω for all $n \in \mathbb{N}$. By Lemma 6, $\mathbb{P}(X | Z)$ admits a version with modulus of continuity ω on $\mathcal{Z}$, hence $\mathbb{P} \in W_\omega$, i.e. W_ω is closed in the weak topology. $\blacksquare$

Lemma 7. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be separable metric spaces. Let $\{\mathbb{P}_n(X | Z)\}_{n \in \mathbb{N}}$ be a family of Markov kernels with shared modulus of continuity ω , and let $\mathbb{P}_n(Y, Z), \mathbb{P}(Y, Z) \in \mathcal{P}(\mathcal{Y} \times \mathcal{Z})$ with $\mathbb{P}_n(Y, Z) \xrightarrow{w} \mathbb{P}(Y, Z)$. Then*

$$d_{BL}(\mathbb{P}_n(X | Z) \otimes \mathbb{P}_n(Y, Z), \mathbb{P}_n(X | Z) \otimes \mathbb{P}(Y, Z)) \rightarrow 0.$$

Proof. Considering the product metric $d_{\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}} = d_{\mathcal{X}} + d_{\mathcal{Y}} + d_{\mathcal{Z}}$, for any $f \in \text{BL}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}; \mathbb{R})$ and $n \in \mathbb{N}$ define $g_n^f : \mathcal{Y} \times \mathcal{Z} \rightarrow \mathbb{R}$ by

$$g_n^f(y, z) := \int_{\mathcal{X}} f(x, y, z) d\mathbb{P}_n(x | z).$$

Then $\|g_n^f\|_\infty \leq 1$, and

$$\begin{aligned} |g_n^f(y, z) - g_n^f(y', z')| &\leq \left| \int_{\mathcal{X}} (f(x, y, z) - f(x, y', z')) d\mathbb{P}_n(x | z) \right| \\ &\quad + \left| \int_{\mathcal{X}} f(x, y', z') d(\mathbb{P}_n(x | z) - \mathbb{P}_n(x | z')) \right| \\ &\leq d_{\mathcal{Y}}(y, y') + d_{\mathcal{Z}}(z, z') + d_{BL}(\mathbb{P}_n(X | z), \mathbb{P}_n(X | z')) \\ &\leq d_{\mathcal{Y}}(y, y') + d_{\mathcal{Z}}(z, z') + \omega(d_{\mathcal{Z}}(z, z')) \end{aligned}$$

by the bounded Lipschitz property of f and the shared modulus of continuity of $\{\mathbb{P}_n(X | Z)\}$. Hence the family $\mathcal{G} := \{g_n^f : n \in \mathbb{N}, f \in \text{BL}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}; \mathbb{R})\}$ is uniformly bounded by 1 and equicontinuous on $\mathcal{Y} \times \mathcal{Z}$ with shared modulus of continuity.

By Fubini's theorem, we have

$$\begin{aligned} &d_{BL}(\mathbb{P}_n(X | Z) \otimes \mathbb{P}_n(Y, Z), \mathbb{P}_n(X | Z) \otimes \mathbb{P}(Y, Z)) \\ &= \sup_f \int f d(\mathbb{P}_n(X | Z) \otimes \mathbb{P}_n(Y, Z) - \mathbb{P}_n(X | Z) \otimes \mathbb{P}(Y, Z)) \\ &= \sup_f \int g_n^f d(\mathbb{P}_n(Y, Z) - \mathbb{P}(Y, Z)) \\ &\leq \sup_{g \in \mathcal{G}} \int g d(\mathbb{P}_n(Y, Z) - \mathbb{P}(Y, Z)). \end{aligned}$$

Since $\mathbb{P}_n(Y, Z) \xrightarrow{w} \mathbb{P}(Y, Z)$, this last term converges to 0 by Dudley (2002), Corollary 11.3.4. $\blacksquare$

Lemma 8. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be separable metric spaces, with $\mathcal{Z}$ complete. If $\mathbb{P}_n(X | Z) \rightarrow \mathbb{P}(X | Z)$ uniformly on compacta in d_{BL} , then for any given $\mathbb{P}(Y, Z)$ we have*

$$d_{BL}(\mathbb{P}_n(X | Z) \otimes \mathbb{P}(Y, Z), \mathbb{P}(X | Z) \otimes \mathbb{P}(Y, Z)) \rightarrow 0.$$

Proof. For any $f(x, y, z) \in \text{BL}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}; \mathbb{R})$ we have that $x \mapsto f(x, y, z) \in \text{BL}(\mathcal{X}; \mathbb{R})$, so

$$\left| \int f(x, y, z) d\mathbb{P}_n(x | z) - \int f(x, y, z) d\mathbb{P}(x | z) \right| \leq d_{BL}(\mathbb{P}_n(X | z), \mathbb{P}(X | z)).$$

Since $\mathcal{Z}$ is separable and complete, the measure $\mathbb{P}(Z)$ is tight, so for every $\varepsilon > 0$ there exists a compact $K_\varepsilon \subseteq \mathcal{Z}$ such that $\mathbb{P}(Z \notin K_\varepsilon) \leq \varepsilon$. Since $d_{BL} \leq 2$, we have

$$\begin{aligned} & \left| \int f(x, y, z) d\mathbb{P}_n(x|z) d\mathbb{P}(y, z) - \int f(x, y, z) d\mathbb{P}(x|z) d\mathbb{P}(y, z) \right| \\ & \leq \int d_{BL}(\mathbb{P}_n(X|z), \mathbb{P}(X|z)) d\mathbb{P}(z) \\ & \leq \sup_{z \in K_\varepsilon} d_{BL}(\mathbb{P}_n(X|z), \mathbb{P}(X|z)) + 2\varepsilon. \end{aligned}$$

Since $\mathbb{P}_n(X|Z) \rightarrow \mathbb{P}(X|Z)$ uniformly on compacta in d_{BL} we get the result. $\blacksquare$

Theorem 9. *Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be complete separable metric spaces with $\mathcal{Z}$ locally compact. If $\mathbb{P}_n(X, Y, Z) \xrightarrow{w} \mathbb{P}(X, Y, Z)$ where $X \perp_{\mathbb{P}_n} Y|Z$ and $\mathbb{P}_n(X|Z)$ has modulus of continuity ω for all $n \in \mathbb{N}$, then $X \perp_{\mathbb{P}} Y|Z$.*

Proof. By Lemma 6, $\mathbb{P}_n(X|Z) \rightarrow \mathbb{P}(X|Z)$ uniformly on compact subsets of $\mathcal{Z}$ in d_{BL} . By the chain inequality

$$\begin{aligned} & d_{BL}(\mathbb{P}_n(X|Z) \otimes \mathbb{P}_n(Y, Z), \mathbb{P}(X|Z) \otimes \mathbb{P}(Y, Z)) \\ & \leq d_{BL}(\mathbb{P}_n(X|Z) \otimes \mathbb{P}_n(Y, Z), \mathbb{P}_n(X|Z) \otimes \mathbb{P}(Y, Z)) \\ & \quad + d_{BL}(\mathbb{P}_n(X|Z) \otimes \mathbb{P}(Y, Z), \mathbb{P}(X|Z) \otimes \mathbb{P}(Y, Z)), \end{aligned}$$

the first term converges to zero by Lemma 7 (since $\mathbb{P}_n(Y, Z) \xrightarrow{w} \mathbb{P}(Y, Z)$, which follows from $\mathbb{P}_n(X, Y, Z) \xrightarrow{w} \mathbb{P}(X, Y, Z)$), and the second term converges to zero by Lemma 8. Hence $\mathbb{P}_n(X|Z) \otimes \mathbb{P}_n(Y, Z) \xrightarrow{w} \mathbb{P}(X|Z) \otimes \mathbb{P}(Y, Z)$. Combined with $\mathbb{P}_n(X, Y, Z) = \mathbb{P}_n(X|Z) \otimes \mathbb{P}_n(Y, Z)$ and $\mathbb{P}_n(X, Y, Z) \xrightarrow{w} \mathbb{P}(X, Y, Z)$, this gives $\mathbb{P}(X, Y, Z) = \mathbb{P}(X|Z) \otimes \mathbb{P}(Y, Z)$, i.e. $X \perp_{\mathbb{P}} Y|Z$. $\blacksquare$

Acknowledgements

Philip Boeken was supported by Booking.com. Eduardo Skapinakis was funded by national funds through the FCT – Fundação para a Ciência e a Tecnologia, I.P., under the scope of the projects UIDB/00297/2020 (<https://doi.org/10.54499/UIDB/00297/2020>) and UIDP/00297/2020 (<https://doi.org/10.54499/UIDP/00297/2020>) (Center for Mathematics and Applications) and the FCT scholarship, reference 2022.10596.BD (<https://doi.org/10.54499/2022.10596.BD>). The authors thank Claude Code, Bas Kleijn, Sourbh Bhadane and Johannes Ruf for helpful remarks and suggestions.

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